

اتحاد طلبة هندسة الطاقة والاتصالات

PCE

الطبعة الثانية

مختبر الكترونيات

Dont miss the change.. be part of it



Groups/pce.fet



PCE_FET

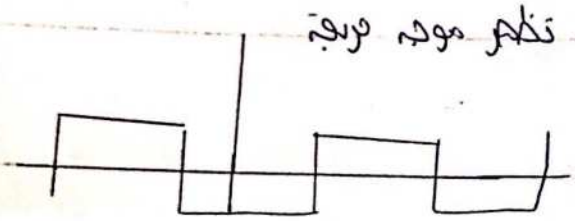
Scanned with CamScanner

experiment -1- : Introduction.

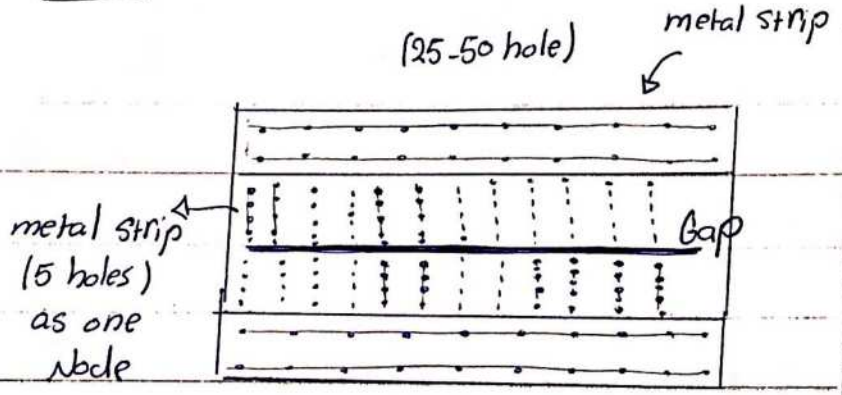
Equipments and materials :-

1. Dc power supply
2. function generator
3. osilloscope
4. BNC probe

عند فحص البروب اذا كان سالم فيجب أن تظهر موجه مربعة



5. Bread Board

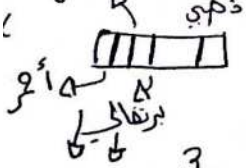


6. Resistors

- potentiometer : (vairable resistor
- fixed carbon Resistors

كود قيم المقاومة عن طريق الالوان

لحساب قيم المقاومة نتعرف كارهم بـ 10 لاتزلون لتحديد نسبة الخطأ.



من اليسار للمين كانت الالوان أحمر 2 أخضر 5 برتقالي 3

$25 \times 10^3 \pm 5\%$

$25 \times 10^3 \pm 5\%$ قيم المقاومة

نسبة الخطأ

Scanned with CamScanner

capacitors

Polarized capacitors: large value 1μf

They must connected correctly

- يجب ان يتم وصل المواسع مع الدارة تبادا على قطبيته

- Un - polarized capacitors

- لحساب مواسعة المواسع

لـ الرقم الأول هاهو والثاني كذلك , نضرب بـ 10 أس الرقم الثالث

والوحدة pf

Ex: 102 has painted on a small value capacitor, determine its value in f ?

Sol

1st digit = 1 , 2nd digit = 0 , 3rd digit is multiplier = 10^2

Then $C = 10 * 10^2$ pf = 1000pf

Black	0		
Brown	1		
Red	2	Gold	±5%
orange	3	selver	±10%
yellow	4	None	±20%
Green	5		
Blue	6		
Violt	7		
Gray	8		
White	9		

* الألوان التي تمثل المقاومات

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Diode type ?

1. light Emitting Diode (LED)

this diode permits the transfer of electric current between the electrodes, light is produced

2. Avalancue diode

operates in the reverse bias and used valncue effect for its operation

3. laser diode ..

it praduce coherent light

4- Zener Diode ..

(Regulator it provides a stable refrece voltage

5* small signal or small current diode → these diode assumes that the operating point is not affected becuase the signal small

6- large signal diodes → the operating point in these diodes get affected as the signal is large

7- Transient voltage suppression diodes → this diode is useal to protect the electronic that are sensitive against voltage spikes

8- point contact diode

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9. photo diode → it used to delect light and feature wide, trasparent junction

10. Varicap Diode (Varactor) → This type of diode feature a reverse bias placed upon it

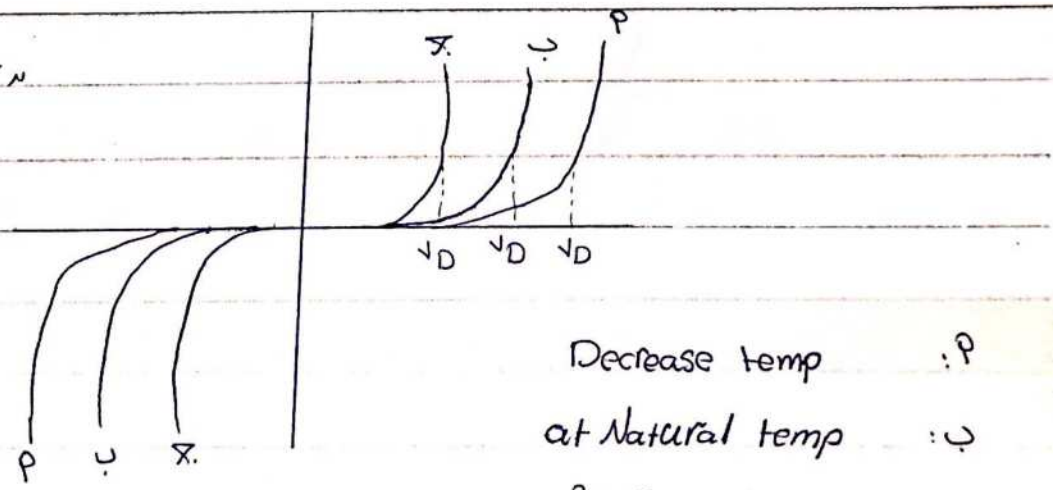
11. Rectifier Diode → retify alternating power

12. Silicon controlled rectifier → triggered to the oN condition due to the application of small voltage

13. Vacum diodes → can tolerate high inverse voltage

تأثير تيار الـ Diods بدرجة الحرارة ← كلما زادت درجة الحرارة تجيح الالكترونات أسرع وتمتلك طاقة أكبر وبالتالي تقل V_D والعكس صحيح

„ temp Effect „



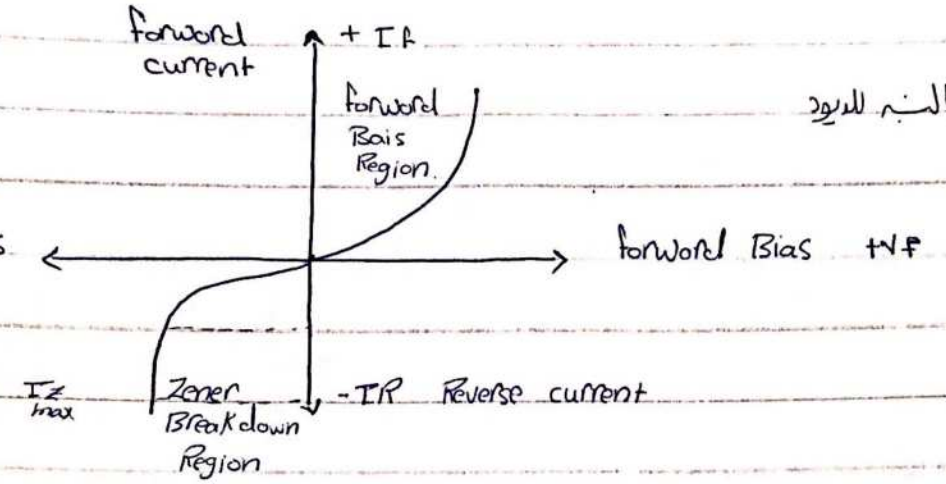
ج : Decrease temp

ب : at Natural temp

أ : Increase temp

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experiment one : (General purpose diode)



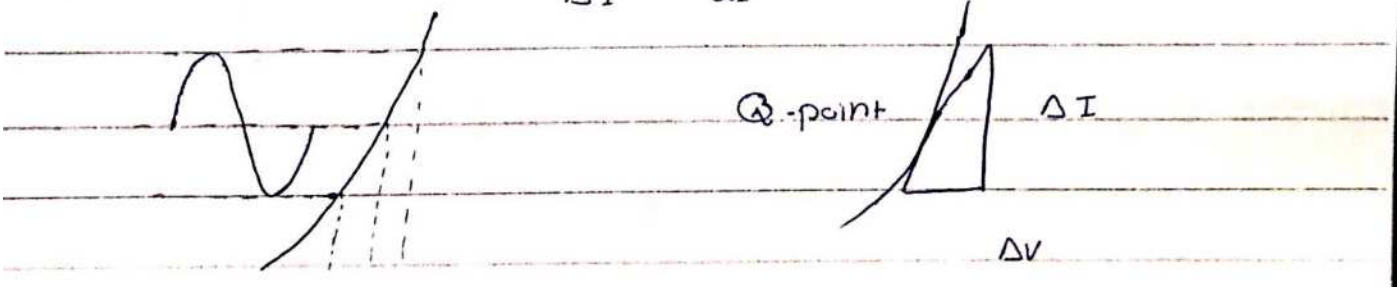
Diode I-V charateristics

if dc voltage is applied, then the type of resistance is static

$$R_S = \frac{V_D}{I_D}$$

if an Ac voltage is applied the resistance is dynami

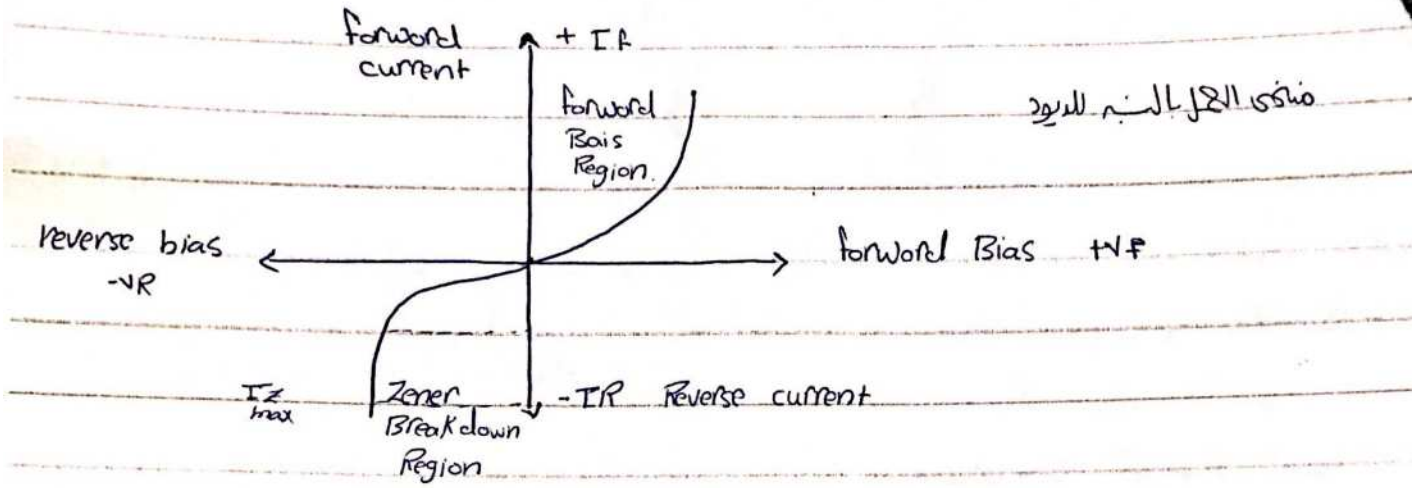
$$r_d = \frac{\Delta V}{\Delta I} = \frac{dV}{dI}$$



(A) (B)

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experiment one : (General purpose diode)



منحنى العمل الخاص للديود

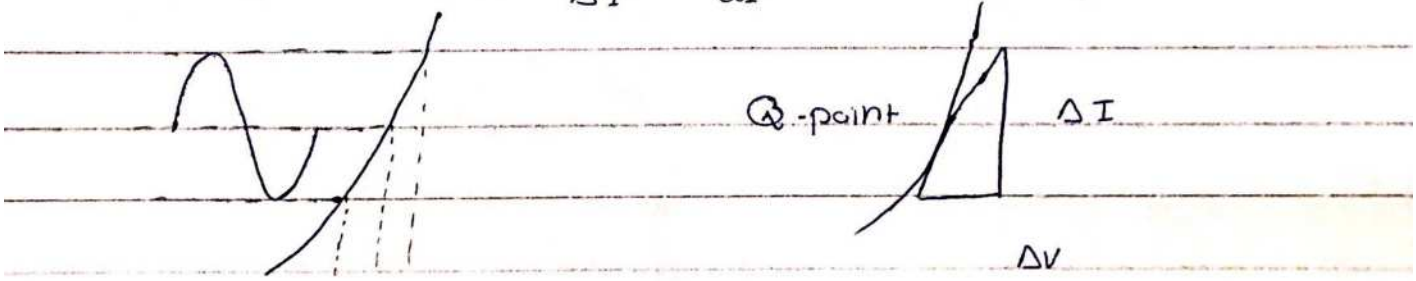
Diode I-V charateristics

if dc voltage is applied, then the type of resistance is static

$$R_S = \frac{V_D}{I_D}$$

if an Ac voltage is applied the resistance is dynami

$$r_d = \frac{\Delta V}{\Delta I} = \frac{dv}{dI}$$



(A) (B)

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part one

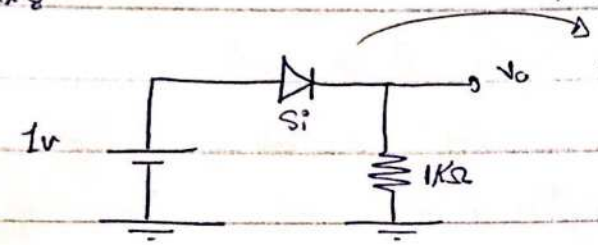
- يمتلك الديود فولتيه تسمى V_D وتساوي

Si → 0.7

Ge → 0.3

ideal → ∅

Ex 8



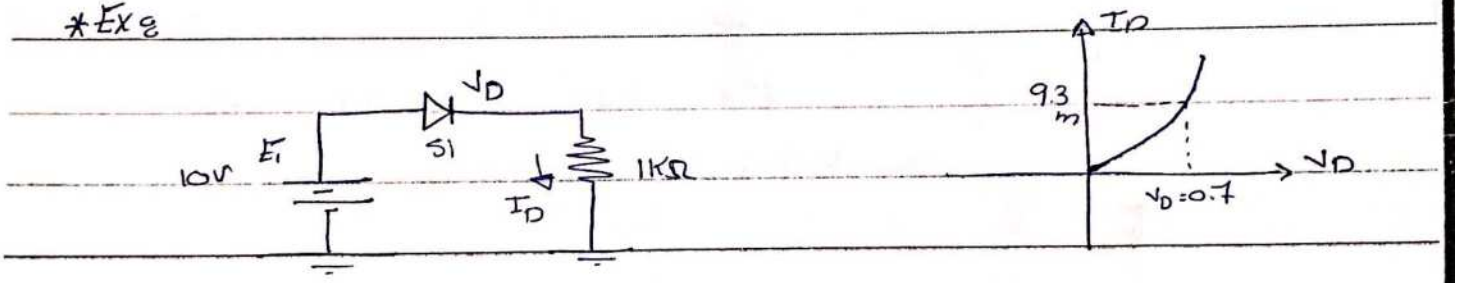
- يجب أن تكون الفولتيه الداخله الى الديود أكبر من فولتية V_D حتى يعمل الديود والشرط الثاني ان يكون الديود موصول forward bias

find I_D, V_D

$$-1 + 0.7 + 1K * I_D = 0 \rightarrow I_D = \frac{0.3}{1K} = 3mA$$

$$V_o = 1K * I_D = 0.3V$$

*Ex 8



to find I_D ?

$$-E_s + V_D + R_L I_D = 0$$

$$-10 + 0.7 + 1K I_D = 0 \rightarrow I_D = \frac{10 - 0.7}{1K} = 9.3mA$$

$$V_R = R_L I_D = 1K * 9.3mA = 9.3V$$

- To find the load line Form the eqn ?

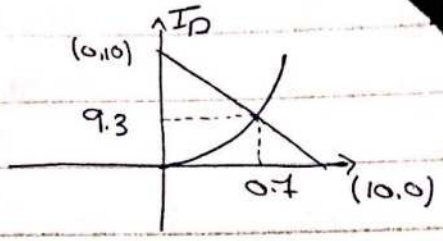
$$-E_s + V_D + R_L I_D = 0 \qquad V_D = 0$$

$$-10 + 1K * I_D \rightarrow I_D = 10mA$$

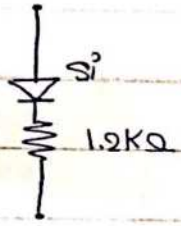
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$-E_s + V_{D1} + R_2 I_D = 0$ — $I_D = 0$

$-10 + V_D \rightarrow V_D = 10\,Volt$



*Example



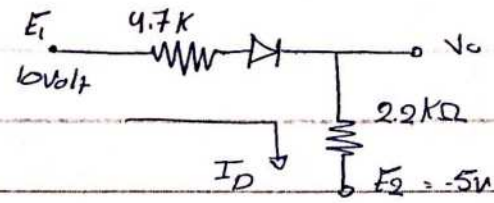
find I_D ?

$V_D = 0.7 > 0.5$

The Diode is off (o.c)

I_D = Zero

[2]



find V_o ?

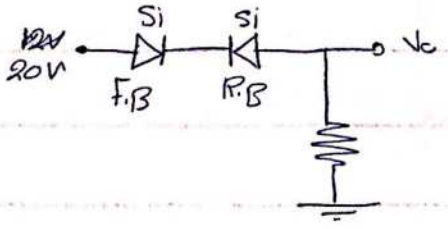
$-10 + 4.7K I_D + 0.7 + 2.2K I_D + 5$

$$I_D = \frac{10+5-0.7}{4.7K+2.2K} = 2.07\,mA$$

$V_o = V_R + E_2$

$= 2.2K * 2.07m + 5 = -0.446\,Volt$

[3]

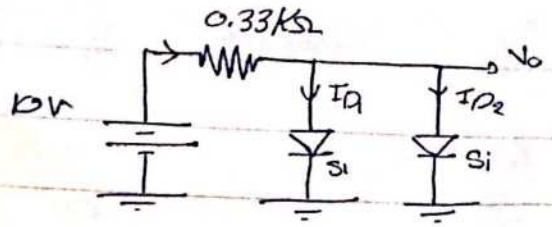


I_D = Zero

V_o = Zero

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[4]



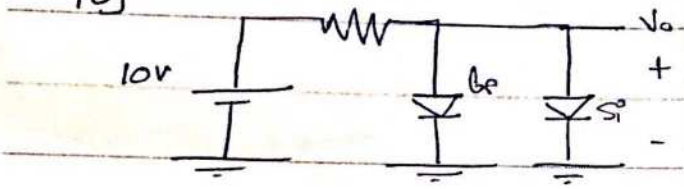
$-10 + 0.33K\,I + 0.7 = 0$

$I = 28.18\,mA$

$I_{D1} = I_{D2} = \frac{I}{2} = 14.09\,mA$

$V_o = 0.7\,Volt$

[5]

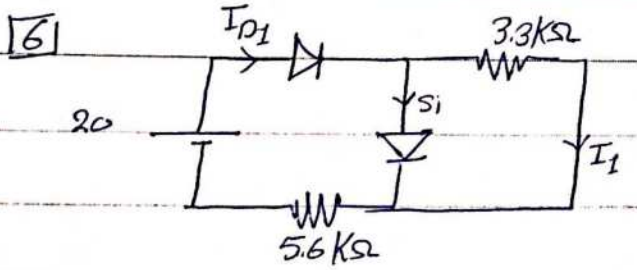


نختار الديود الأقل فولتية

$V_o = V_{D_{Ge}} = 0.3V$

$I_D = 29.39\,mA$

[6]



find I_D , I_{D1} , I_{D2}

$I_1 = \frac{0.7}{3.3K} = 0.21\,mA$

$-20 + 0.7 + 0.7 + 5.6K\,I_D = 0$

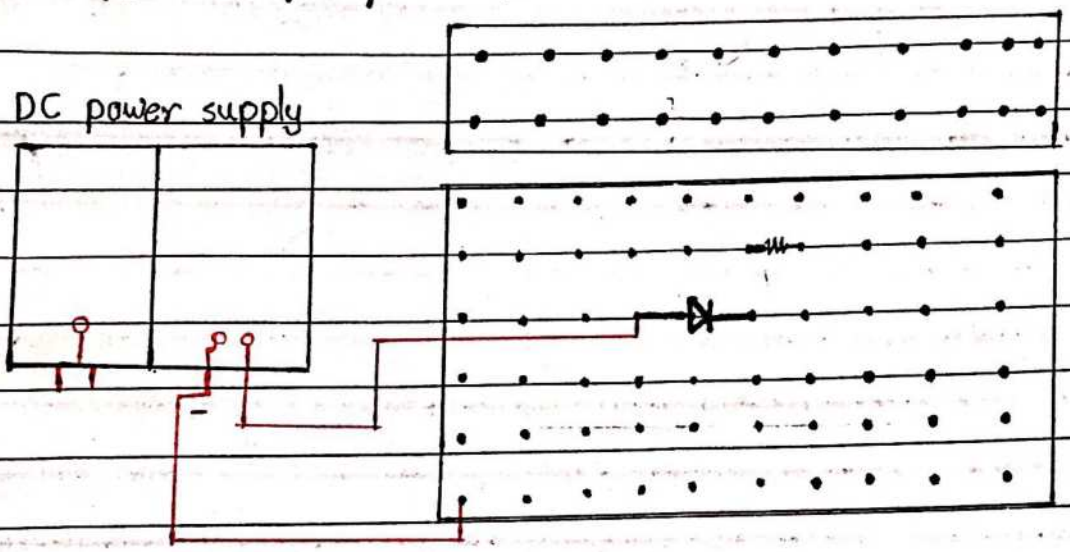
$I_{D1} = 3.32\,mA$

$I_{D2} = I_D - I_1 = 3.1\,mA$

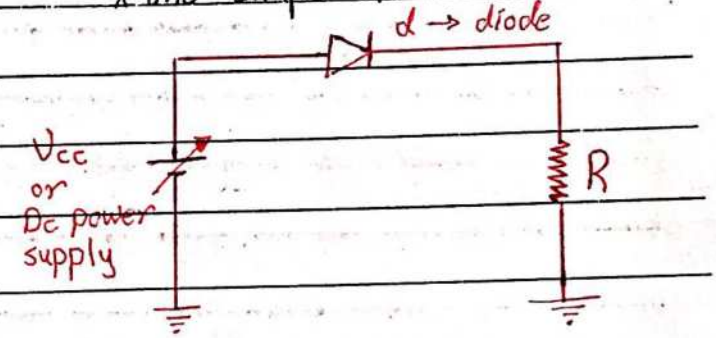
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* Connection of lab electronics *

1 The general purpose Diode :



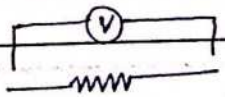
* the shape of connection



* DMM types :

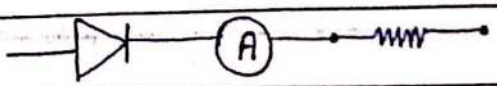
① when need mesure voltage

يشبك مع المقاومة على التوازي



② when need mesure current

يشبك مع المقاومة على التوالي



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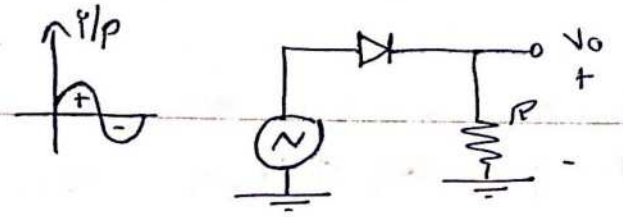
Experiment -2-

Diode Application , Rectifier circuits

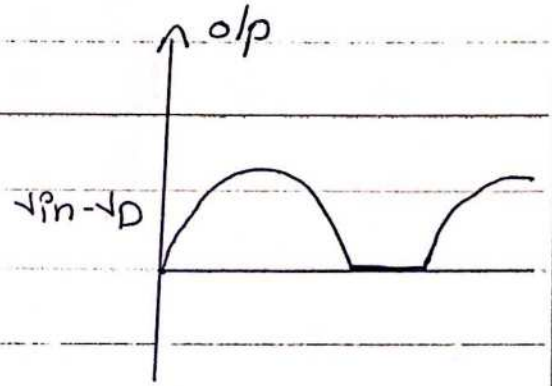
* Ac Application ex

- Rectification
- filter
- limiter (clipper)
- clamper
- Voltage multiplyer

[1] Half Wave Rectifier



- if we applay (+ve) cycle
  Diode ⇒ forword
  $V_o = V_R = V_{in} - V_D$
- if we apply (-ve) cycle
  Diode ⇒ reverse (o.c)
  $V_o = V_R = zero$



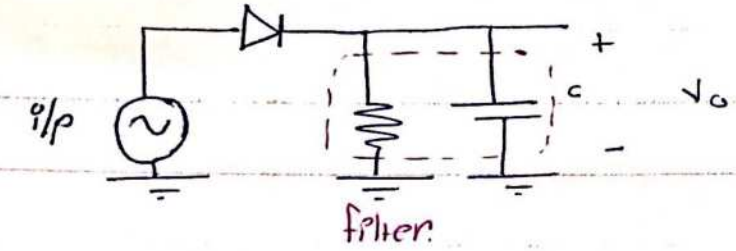
* $f_o = f_{in}$

$V_{DC} = \frac{V_{p(out)}}{\pi}$, $V_p = \sqrt{2}\, V_{rms}$

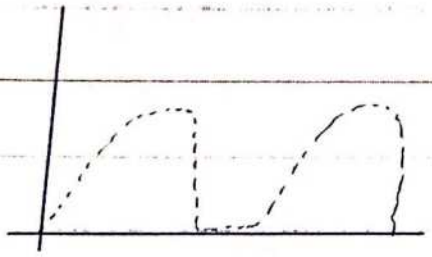
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[2] Half wave Rectifier with capacitor (filter)

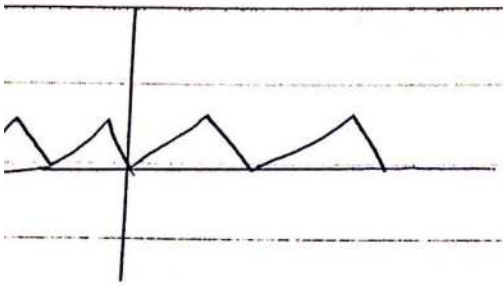
we use filter to
- elemenate the ripple in o/p
- produce a constant level dc



at +ve cycle the capacitor charge يشحن
at -ve cycle the capacitor discharge يفرغ



← signal without capacitor



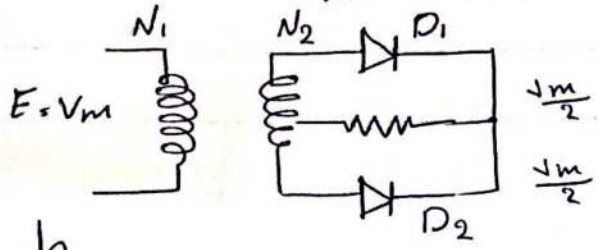
with capacitor

$$V_{r(p-p)} = \frac{1}{f_o \cdot R_L \cdot C} V_{p(out)} \quad , \text{ripple factor} = \frac{V_{r\,p-p}}{V_{DC}}$$

$$V_{DC} = \left[1 - \frac{1}{2 f_o R_L \cdot C}\right] V_{p(out)}$$

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[3] center - Tapped full wave rectifier



if we applay +ve cycle

D_1 ON

D_2 off $\quad V_o = \frac{V_m}{2} - V_{D1}$

if we applay -ve cycle

D_1 off

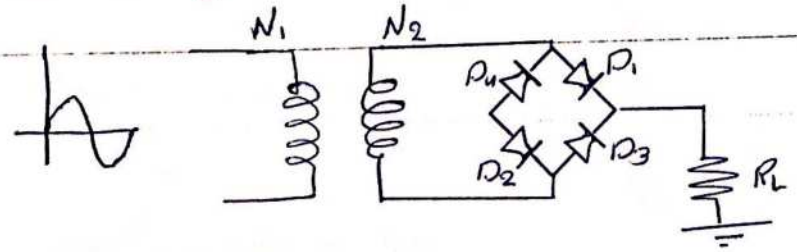
D_2 ON $\quad V_o = \frac{V_m}{2} - V_{D2}$

* في حال اضافة مواسع [filter] نفس قوانين صفحة ٢٤ بالظبط

$F_{out} = 2 f_{in}$

$V_{DC} = \frac{2 V_{P(out)}}{\pi}$

[4] Bridge full wave rectifier



$\frac{N_1}{N_2} = \frac{V_1}{V_2}$

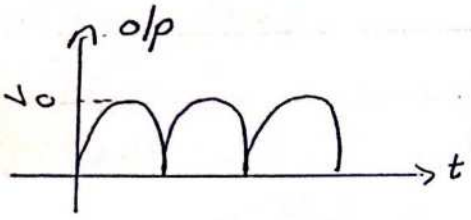
if we applay +ve cycle $D_1, D_2 \rightarrow$ ON , $D_3, D_4 \rightarrow$ off

$V_o = V_m - 2V_D$

Scanned with CamScanner

if we applay -ve cycle → D_1, D_2 → off , D_3, D_4 → ON

$V_o = V_m - 2V_D$



$f_{out} = 2 f_{in}$

$V_{DC} = \frac{2V_{p(out)}}{\pi}$

Rectifiers useful formuls

Parameter	half_wave	center_tap	Bridge
$V_{p(out)}$	$V_p = V_{in} - V_B$	$V_{p(out)} = V_s - V_B$	$V_{p(out)} = V - 2V_B$
output frequncy f_{out}	$f_{out} = f_{in}$	$f_{out} = 2 f_{in}$	$f_{out} = 2 f_{in}$
V_{dc} (without filter)	$V_{DC} = V_{p(out)}/\pi$	$V_{DC} = \frac{2V_{p(out)}}{\pi}$	$V_{DC} = \frac{2V_{p(out)}}{\pi}$
V_{dc} (with filter	$V_{DC} = \left[1 - \left(\frac{1}{2 f_{(out)} R_C}\right)\right] * V_{p(ou}$		
Ripple factor (r)	$r = \frac{V_{r(p-p)}}{V_{DC}}$		
power Dissipation (P_Z)	$P_Z = I_Z V_Z$		
Regulation percentage (%V_o)	%V_o. $[(V_{NL} - V_{FL})/V_{FL}] * 100\%$		

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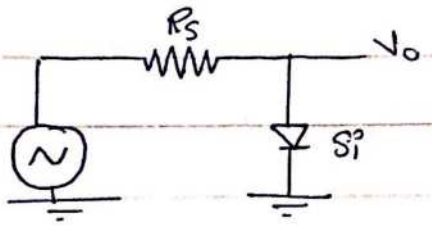
Experiment -3-

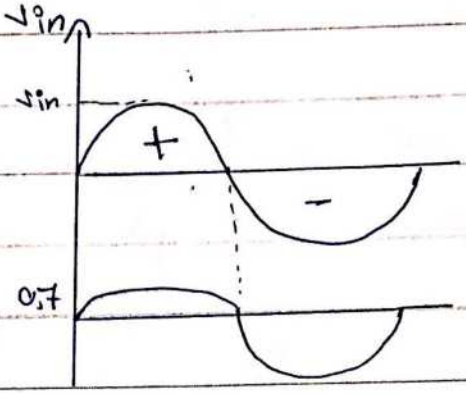
The Diode limiter, the diode clamper.

1 limiter (clipper)

- في حالة الانحياز العكسي يتم تبديل الديود في open.c
- في حالة الانحياز الأمامي يتم تبديل الديود في S.circuit ← اذا كان الديود ideal
  Not ideal ← Dc

Ex *





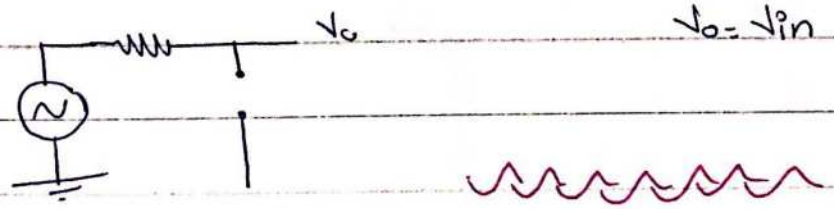
+Ve ← Diode one

بما أن فولتية الكاثود توازي مع الديود ستكون نفس فولتيته

$V_o = V_D = 0.7$ الديود

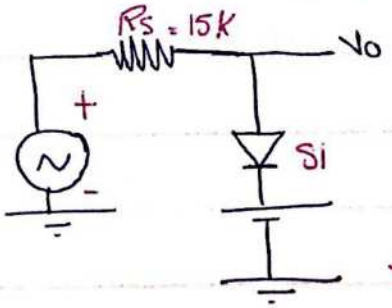
-Ve ← Diode off

open circuit الديود سيكون



ومنها سيكون $V_o = V_{in}$

Ex *

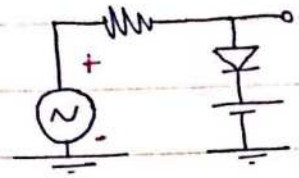


Vin
6Vp-p

$V_{DC} = 1$ Volt

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+ve cycle
Diode on



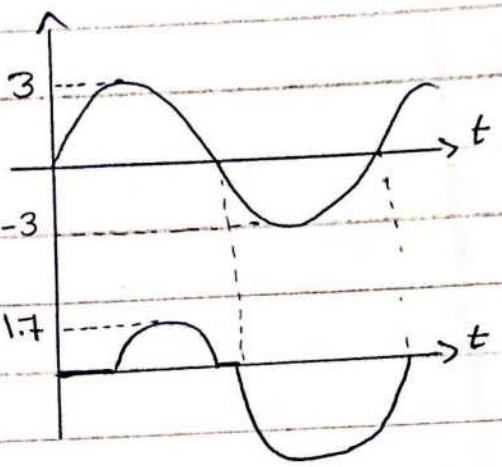
$V_o = 0.7 + 1 = 1.7$

-ve cycle
Diode off Dc وبس مقدر Ac مقدرش

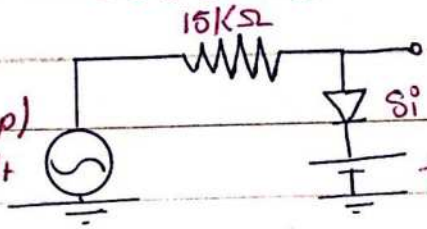
لذا في كل الاحوال سيكون open

$V_o = V_{in}$

* مقدرش Ac وبس مقدر Dc ، يعني فولتية Ac أكبر من فولتية البطارية سيعمل forword Bias



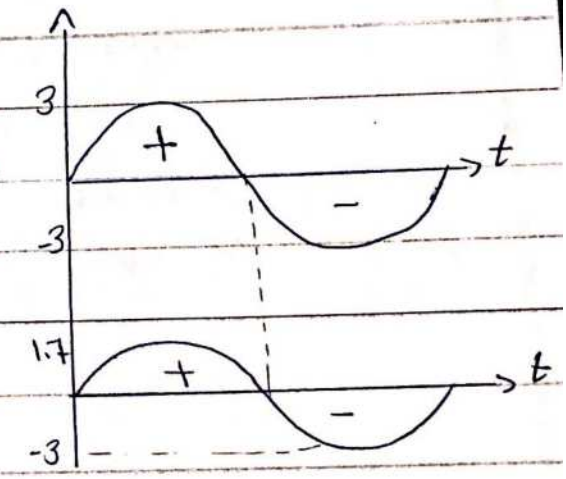
*example 2



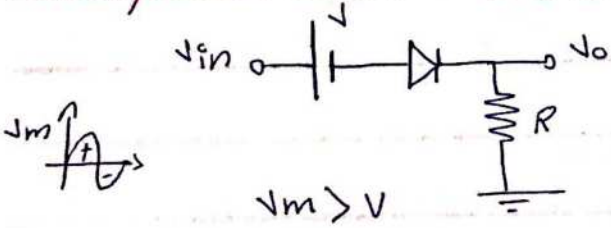
at +ve cycle → $V_o = V_D + 1\,volt$

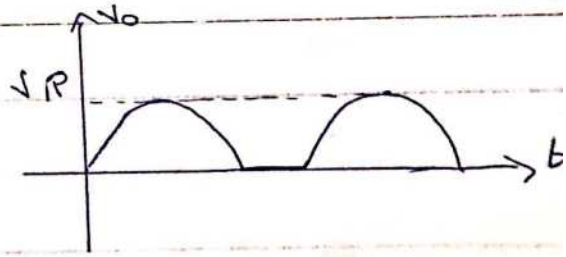
$V_o = 1.7\,volt$

at -ve cycle → $V_o = V_m = 3volt$



example 3-8c





+ve cycle → Diode ON → $V_o = V_m - [V + V_D] = V_R$

-ve cycle → Diode off → V_o = zero , I = zero

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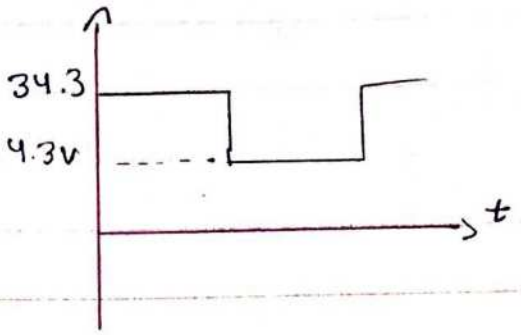
+ve → (Diode off)

$-2.5 - 1.8 + V_o = 0$

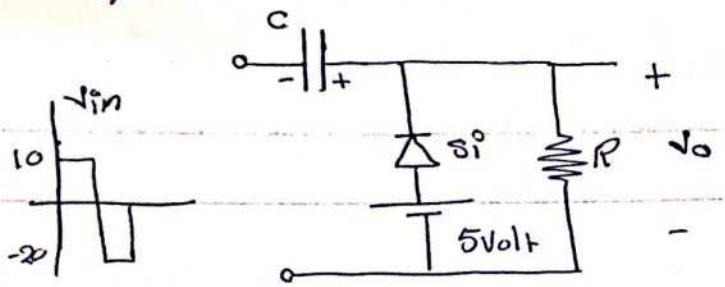
$V_o = 4.3$ Volt

-ve (Diode ON)

$V_o = -0.7$



* example 1 :



[1] -ve cycle

$E - V_C - V_D + 5 = 0$

$20 - V_C - 0.7 + 5 = 0$

$V_C = 24.3$ Volt

$V_o = -V_D + 5 = 4.3$ Volt

[2] +ve cycle

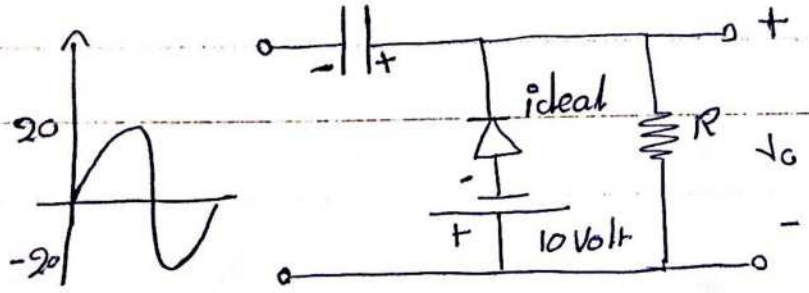
$-E + V_C + V_o = 0$

$V_o = E + V_C$

$= 10 + 24.3$

$= 34.3$ Volt

example 2 :



-ve cycle

$20 - V_C - 10 = 0$

$V_C = 10$ Volt

$V_o = -10$ Volt

+ve cycle

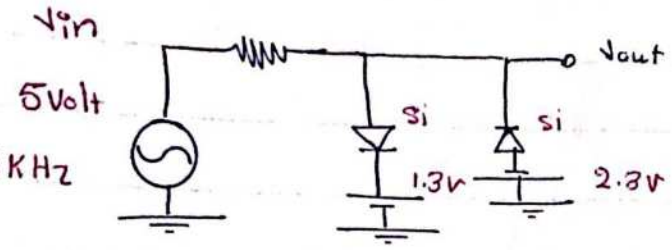
$-20 - V_C + V_o = 0$

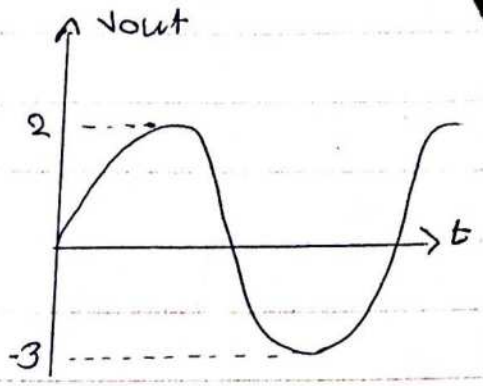
$V_o = 20 + 10$ Volt

$V_o = 30$

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Example 4:





+ve cycle → $V_o = V_D + 1.3 = 2$ Volt

-ve cycle → $V_o = -V_D - 2.3 = -3$ Volt

[2] clamper

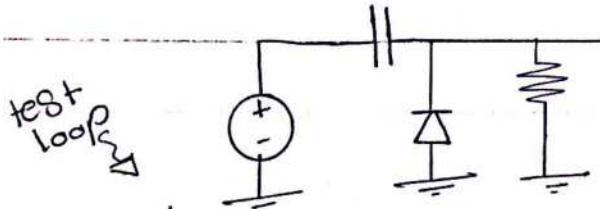
has three component → Diode, resistor, capacitor

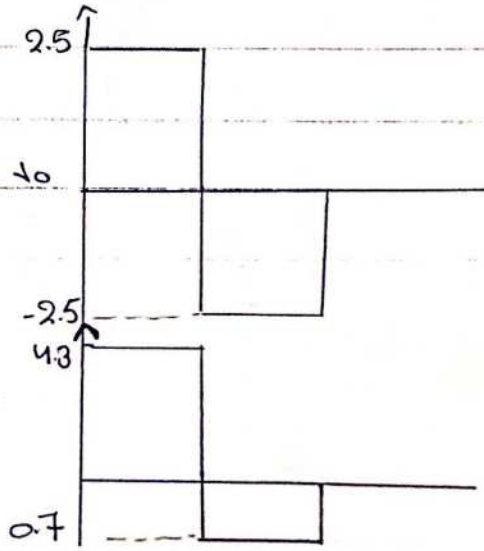
at clamping circuit :-

[1] o/p نفس رسمة i/p فقط ازاحة للأعلى او للأسفل

[2] $V_{(p-p)_{in}} = V_{(p-p)_{out}}$

[3] $\tau = R.C$





نأخذ loop الذي يكون الديود عندما يقبل وهو الجزء السالب

$+2.5 - V_c - 0.7 = 0$

$V_c = 1.8$

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If we apply +ve half (Diode off)

$$-2.5 - 1.8 + V_o = 0$$

$$V_o = 4.3 \text{ V}$$

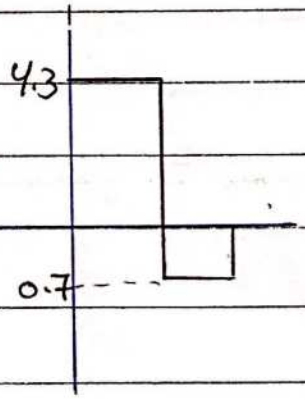
If we apply -ve half (Diode on)

$$V_o = V_D = 0.7$$

$$5 - V - 0.7 = 0$$

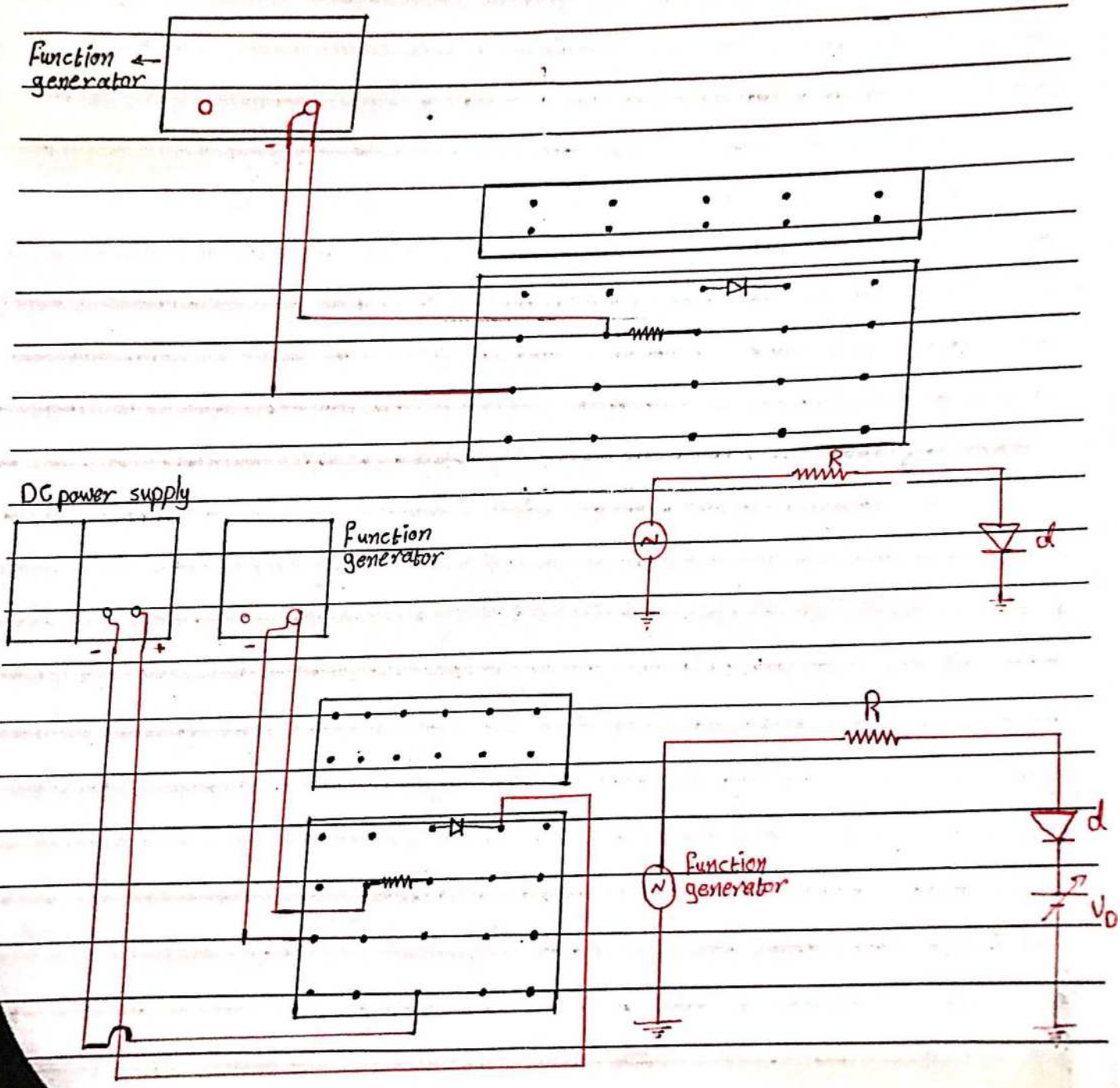
$$V = 4.3$$

$$V_c = 3.3$$

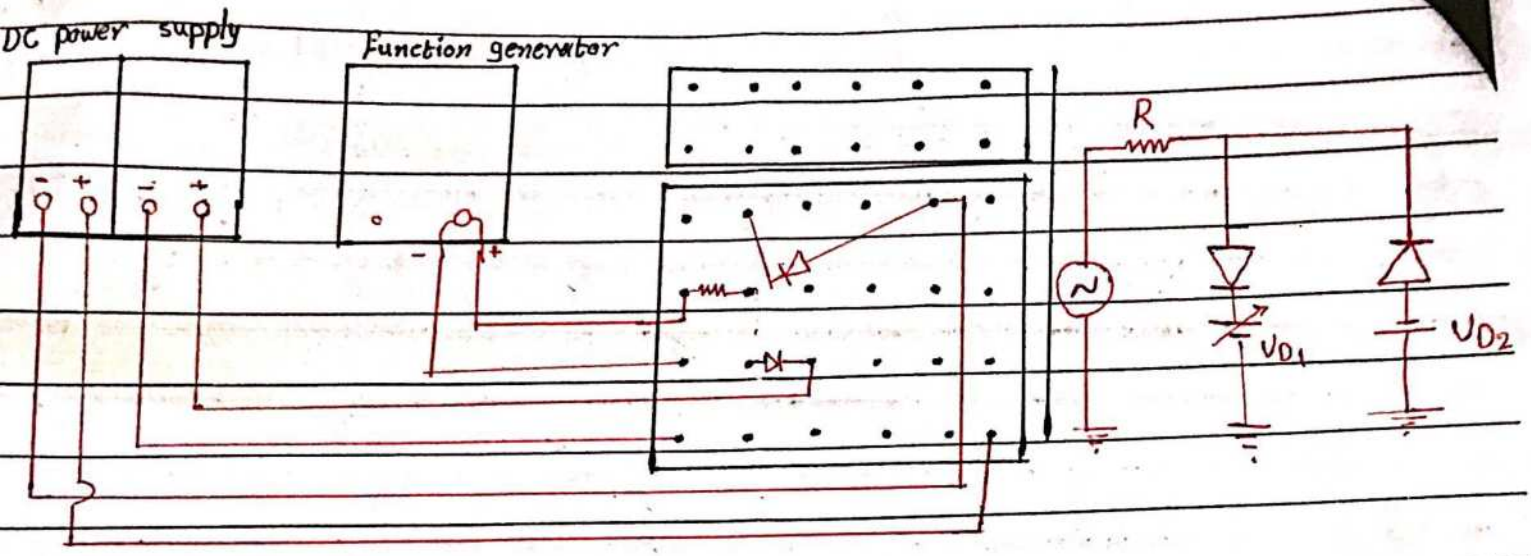


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[2] The diode limiter:



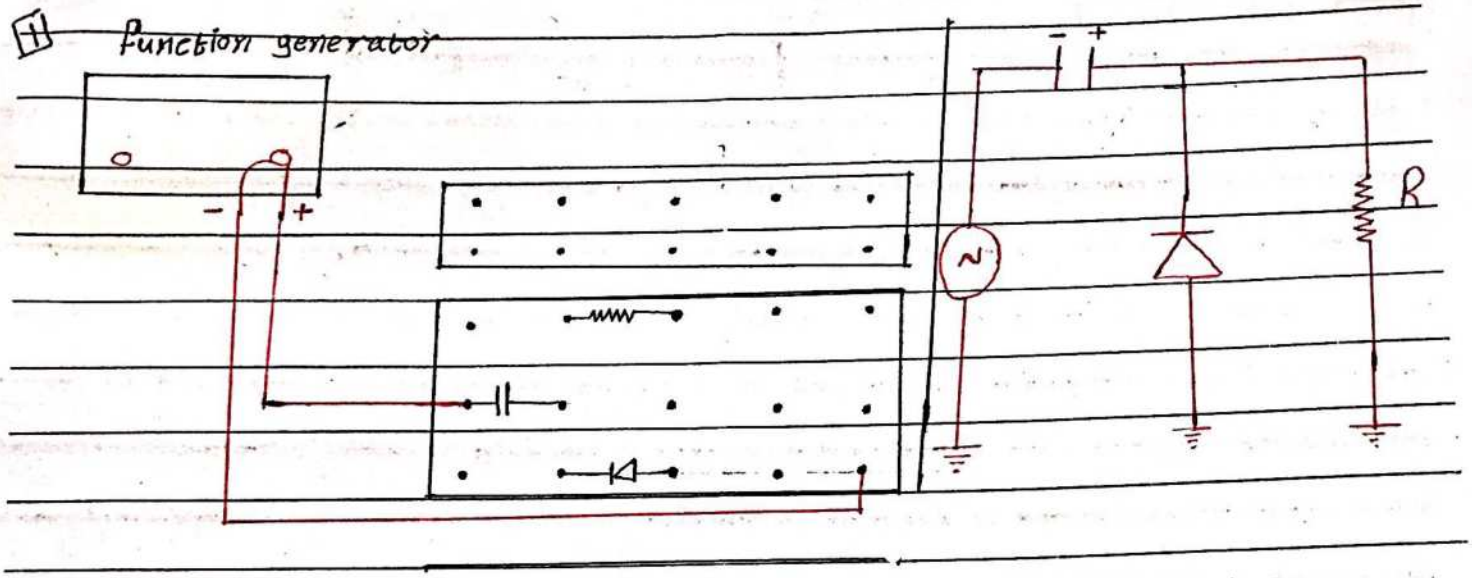
Scanned with CamScanner



Scanned with CamScanner

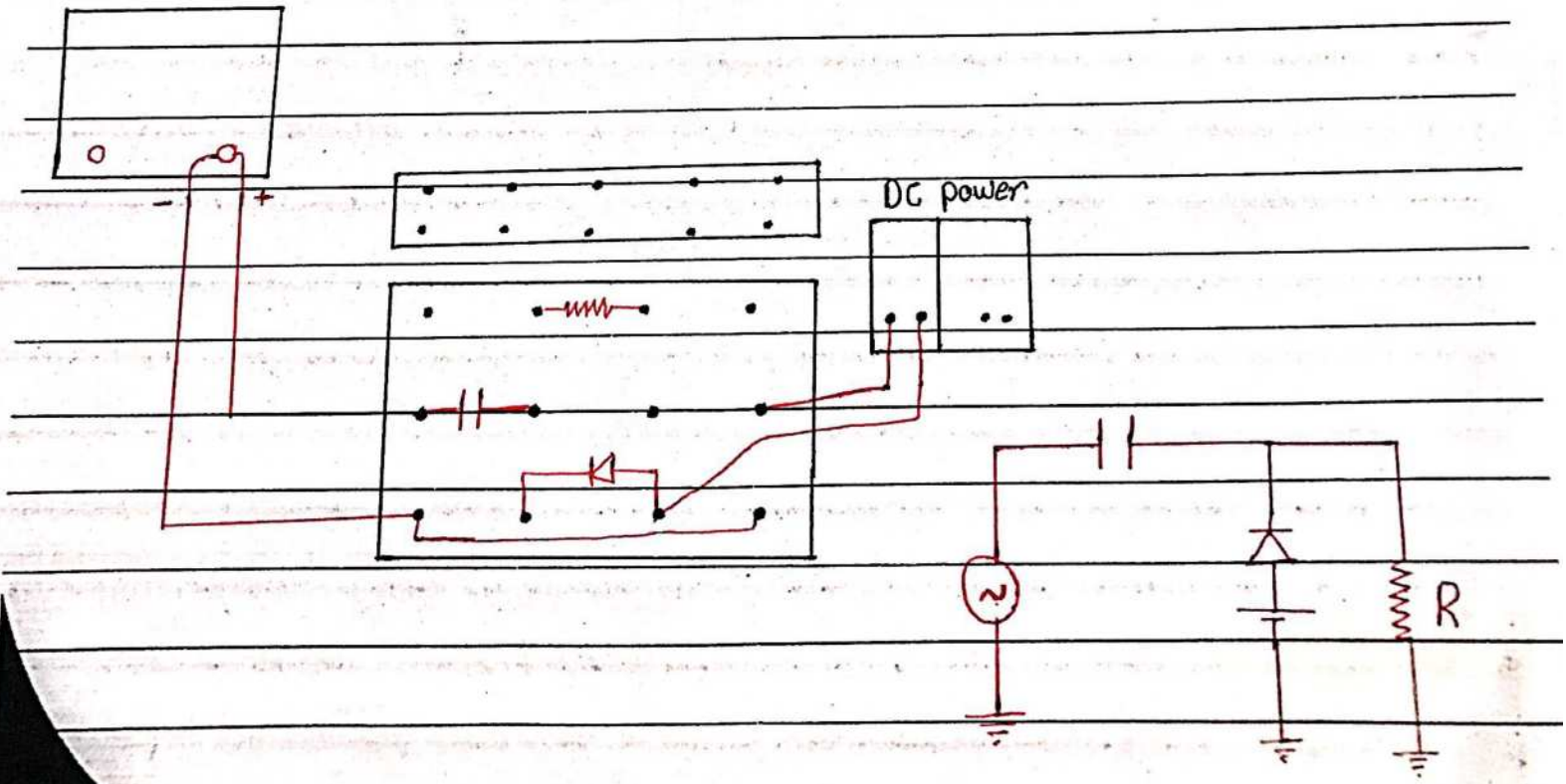
3 The Diode clamper :

1 Function generator



2

Function generator



Scanned with CamScanner

Experiment 4-

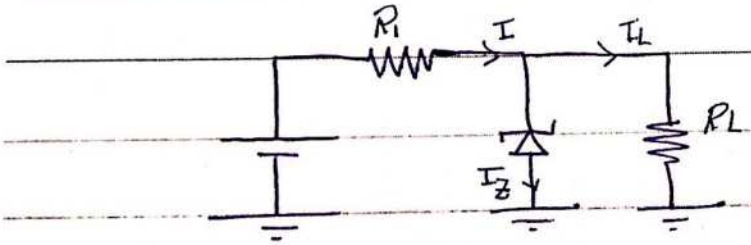
Zener characteristics & Zener As a Voltage regulator

Diode Application two zener as a Voltage regulator

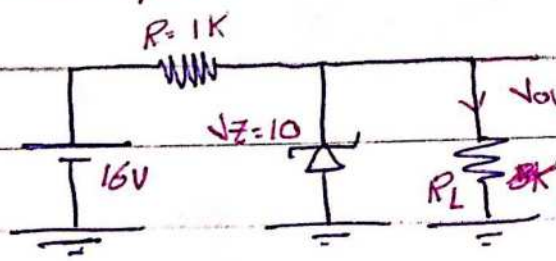
* يعمل الزينر على تنظيم الفولتيه
* يركب ال zener على التوازي مع الحمل حيث يقوم بتثبيت الفولتيه وبالتالي يثبت التيار
* يعمل زينر في الحالتين
  - forward Bios ← كديود عادي ، كمنظم للجهد
  - reverse Bios
* لحل دوائر الزينر اولا نعتبر ال zener (o.c) ونحسب V_{RL}

if $V_{RL} < V_Z$ then Zener (o.c) , $V_o = V_{RL}$

$V_{RL} > V_Z$ then Zener ON , $V_o = V_Z$



Example :-



① $V_{RL} = 16 * \frac{1.2K}{(1.2+1)K} = 8.73 < 10$

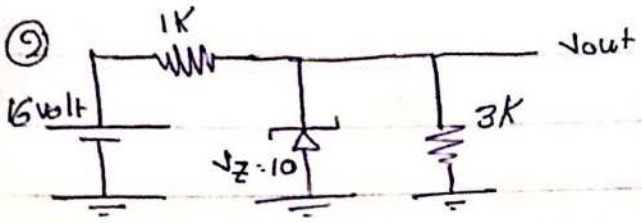
$V_o = V_{RL} = 8.73$ Volt

$I_L = \frac{8.73}{1.2K} = 7.27\,mA = I$

V_Z , I_Z , P_Z = Zero

PCE-FET PCE-FET.COM

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② 

① $V_{RL} = \frac{16*3}{(3+1)} = 12\ Volt > 10$ ⇒ $V_{RL} = V_o = V_z = 10\ Volt$

$V_R = 16 - 10 = 6\ Volt$

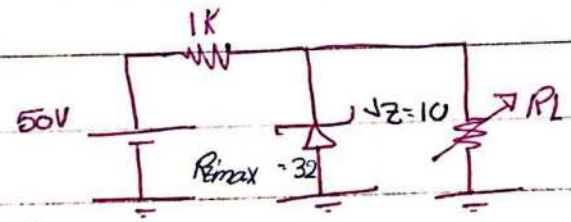
$I = \frac{6}{1K} = 6mA$

$I_L = \frac{V_z}{R_L} = \frac{10}{3K} = 3.33mA$, $I_z = I - I_L$

$6 - 3.33 = 2.67mA$

$P_z = V_z . I_z = 26.7\ mwatt.$

Example



يكون الزينر ON عندما يكون

$V_{RL} = 10Volt$

و الـ R_{min} هي التي تشغل زينر

Find $[R_{Lmin}, R_{Lmax}]$

① $V_{RL} = \frac{V * R_{Lmin}}{(R_{Lmin} + 1K)}$ ⇒ $10 = \frac{50 * R_{Lmin}}{R_{Lmin} + 1K}$

$10[R_{Lmin} + 1K] = 50 R_{Lmin}$

$10R_{Lmin} + 10K = 50 R_{Lmin}$

$R_{Lmin} = \frac{10K}{40} = 250\ \Omega$

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[2] $I_{L\min} = \frac{V}{R_{L\max}}$, $I = I_{Z\max} + I_{L\min}$

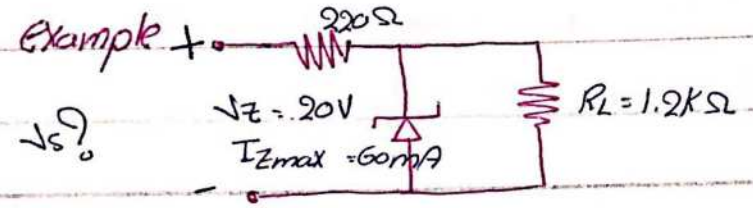
$P_{Z\max} = V_Z I_{Z\max}$ → $I_{Z\max} = \frac{32\,mW}{10V} = 3.2\,mA$

$I = \frac{50-10}{R} = \frac{40}{1K} = 40\,mA$

$I_{L\min} = 40m - 3.2m = 36.8mA$

$R_{L\min} = \frac{10}{36.8m} = 270\,\Omega$ $250 < R_L < 270$

Example +



*find Vs min & Vs max

① $V_R = \frac{V_{s\min} * R_L}{R_L + R}$ - $V_{s\min}$ دائما تكون

$20 = \frac{V_{s\min} * 1.2}{(1.2 + 220)}$ $V_{s\min} = 23.7\,Volt$

② $V_{s\max} = V_R + V_Z$

$= I_{\max} R + V_Z$

$I_{L\max} = I_{Z\max} + I_L$

$= 60m + \frac{20}{1.2K}$ $I_{\max} = 76.67\,mA$

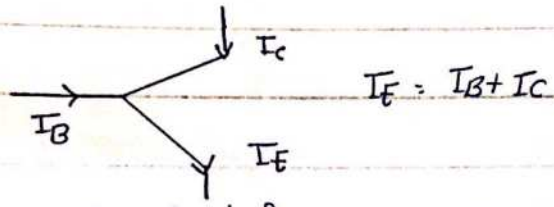
$V_{s\max} = 76.67m * 220\,\Omega + 20 = 36.87\,Volt$

$23.7 < V_s < 36.87$

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Experiment five

Bjt characteristics self-Biased configuration



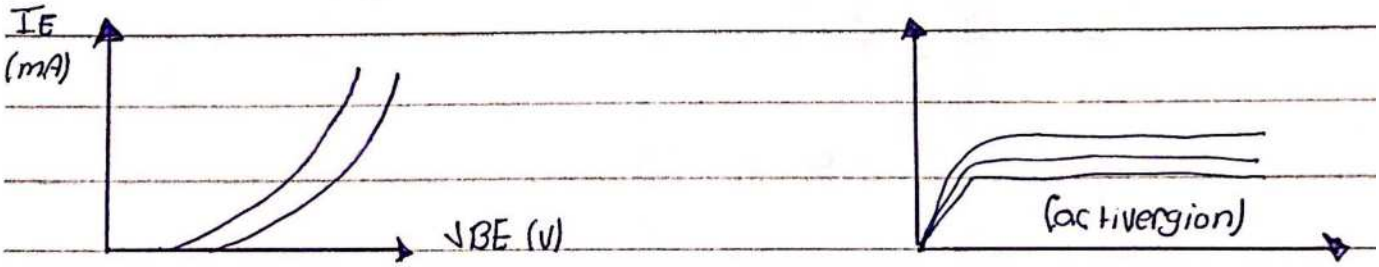
Dc Analysis necessary to establish the proper region of operatic for ac-amplifiction

E.L (Emitter level) → heavily doped.

B.L (Bios level) → lightly doped.

C.L (collecter level) → only lightly doped.

* All calculation applay at npn-transistor, same calculation applay at pnp-transistor, but changing all current direction & voltage polarization.



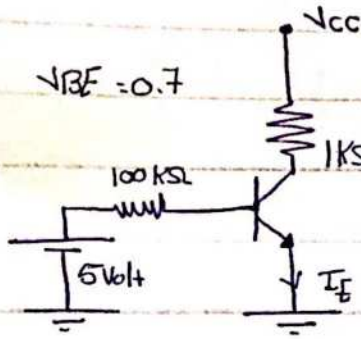
curve for i/p circuit

curve for o/p circuit

"Take Your dreams seriously"

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[1] example: find V_{CE}, I_C, V_{CB} ?



① $-5 + 100K * I_B + V_{BE} = 0$

$I_B = \frac{5 - 0.7}{100K} = 43\mu A$

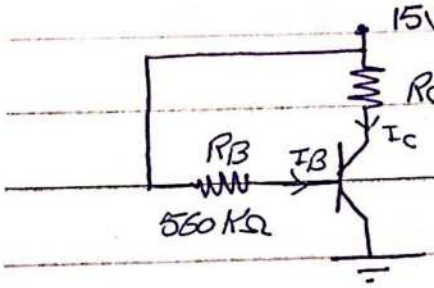
$I_C = \beta I_B = 4.3mA \approx I_E$

② $-15 + 1K\, I_C + V_{CB} + 100K\, I_B + 5 = 0$

$V_{CB} = 10$ Volt

③ $-15 + 1K * I_C + V_{CE} = 0$ $\quad V_{CE} = 10.7$ volt.

[2] Emitter Bios



β = 100

① $-5 + 560K * I_B + 0.7 = 0$

$I_B = 25.5\mu A$

② $I_C = \beta I_B = 2.55$ mA

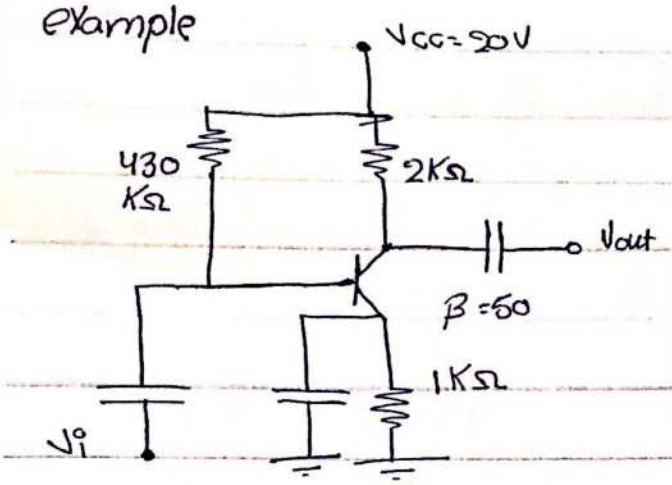
③ $-5 + R_C I_C + V_{CE} = 0$

$V_{CE} = 12.45$ Volt

* العيوب من كل دوائر الـ Bjt زيادة الـ Stability وايضا هي أقل تأثير بـ β
* تعتمد β على درجة الحرارة اعتماد مباشر ، أي تغيير بدرجة الحرارة حيث تغير على الـ β وقيمة الـ I_C

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Example



① $-20 + 430 * I_B + 0.7 + 1K(1+\beta) I_B = 0$

$I_B = 40.1\ \mu A$

$I_C = \beta I_B = 20.01\ mA$

② $V_C = V_{CC} - R_C I_C = 15.98$ Volt

③ $V_E = I_E R_E = (1+\beta) I_B \cdot R_E$

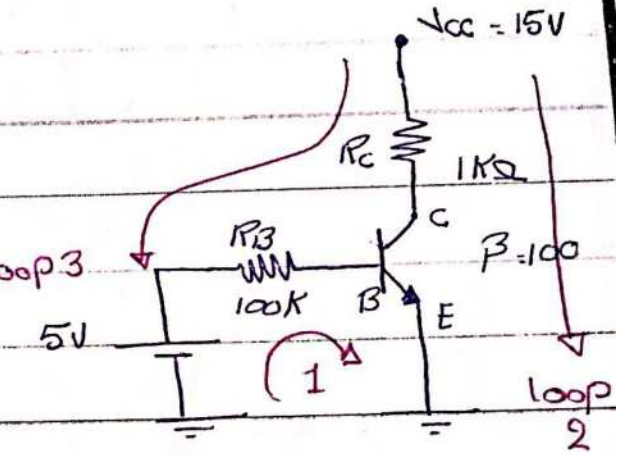
$= 2.05$ Volt

④ $V_B = V_{BE} + V_E = 2.75$ Volt

Example 2 $V_{BE} = 0.7$

$I_C \simeq I_E \simeq \beta I_B$

في الغالب يتم أخذ اللوب الاول لايجاد I_C



for loop 1

$-5 + R_B I_B + V_{BE} = 0$

$I_B = 43\ \mu A$, $I_C = \beta I_B = 4.3\ mA$

for loop 2

$-15 + R_C I_C + V_{CE} = 0 \rightarrow V_{CE} = 11.7$ Volt

loop 3 عكس اتجاه السهم

$-15 + R_C I_C + V_{CB} - R_B I_B + 5 = 0 \rightarrow V_{CB} = 10$ Volt

find V_B → طريقة 1: $V_{BE} = V_B - V_E$ → $0.7 = V_B - 0$ (موجود الارضي مباشرة)

طريقة ثانية ↳ $-5 + I_B R_B + V_B = 0$ $V_B = 0.7$ Volt

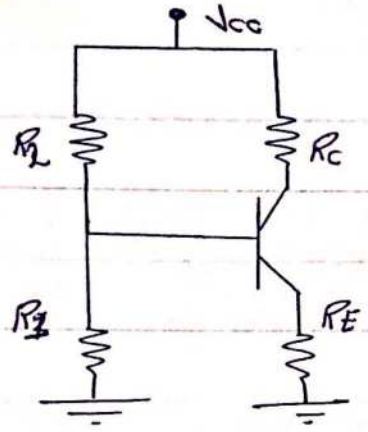
بأخذ loop1 وتاريخي جيب V_B

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Experiment -6-

Voltage divider - Bias configuration common emitter amplifier

II Voltage divider



→ approximate method

$\beta R_E \geq 10 R_2$

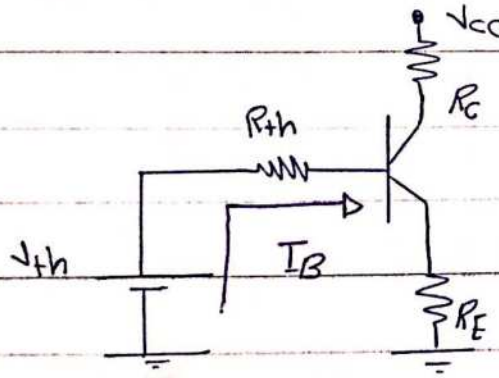
$V_B = V_{cc} \frac{R}{R_1 + R_2}$

$V_{BE} = V_B - V_E$

$I_E = \frac{V_E}{R_E}$, $I_C = \beta I_B$

→ Exact methode

$V_{th} = \frac{V_{cc} R_2}{R_1 + R_2}$, $R_{th} = R_1 // R_2$



$r_e = \frac{26\ mV}{I_E}$

$A_V = \frac{V_o}{V_i}$

$A_V = \frac{R_C // R_L}{R_{E_1} + r_e}$, Normal condition

$A_V = \frac{R_C}{R_{E_1} + r_e}$, No load

$A_V = \frac{R_C // R_L}{R_{E_1} + R_{E_2} + r_e}$, No Bypass capacitor

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Useful formulas

Quiescent dc base voltage

(1) $V_B = \left(\frac{R_2}{R_1 + R_2}\right) V_{CC}$

Quiescent dc emitter voltage

(2) $V_E = V_B - V_{BE}$

Quiescent dc emitter current

(3) $I_E = \frac{V_E}{R_{E1} + R_{E2}}$ $(I_C = I_E)$ for large β

Quiescent dc collector voltage

(4) $V_C = V_{CC} - I_C R_C$

Quiescent dc collector-emitter voltage

(5) $V_{CE} = V_{CC} - I_C (R_C + R_{E1} + R_{E2})$

Voltage gain from base to collector

(1) $A_V = \frac{V_{out}}{V_{in}}$

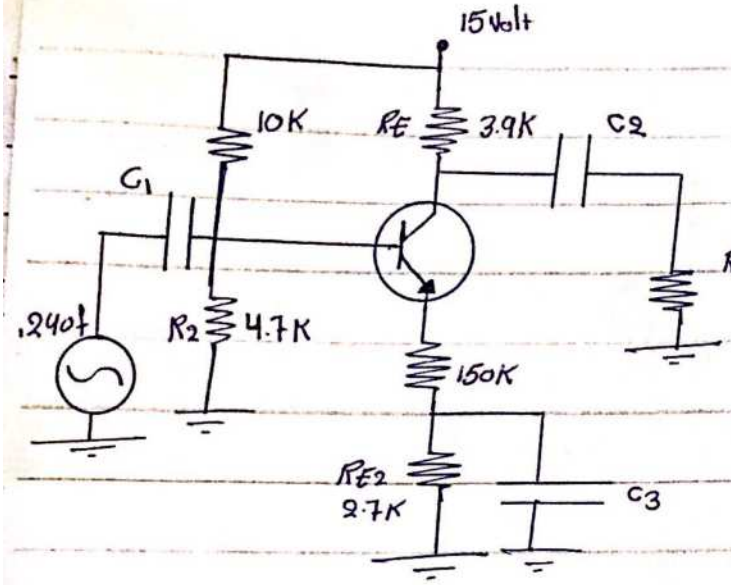
(2) $A_V = (R_C \parallel R_L)/(R_{E1} + r_e)$ (Normal condition)

(3) $A_V = \frac{R_C}{R_{E1} + r_e}$ (No load)

(4) $A_V = \frac{R_C \parallel R_L}{R_{E1} + R_{E2} + r_e}$ (No Bypass capacitor)

Transistor ac resistance (at normal room temperature) $r_e = \frac{26}{}$

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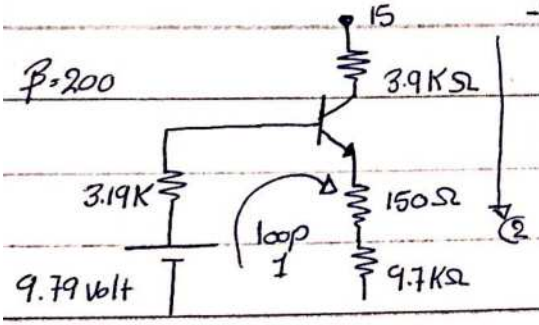


$V_{th} = V_B = \frac{15 * 4.7}{10 + 4.7}$

$V_{th} = 4,7959$ Volt

$R_{th} = \frac{R_1 * R_2}{R_1 + R_2} = 3,578$

$C_1, C_2, C_3 \rightarrow$ open



$-4,79 + 3.57\,I_b + V_{BE} + 285K\,I_E = 0$

$-4.79 + 3.57K\,I_b + 0.7 + 285K\beta I_B = 0$

$I_B = 7.13\mu A$

$I_E = \beta I_B = 1.426mA$

Ac Analysis.

$r_e = \frac{26mV}{I_E}$

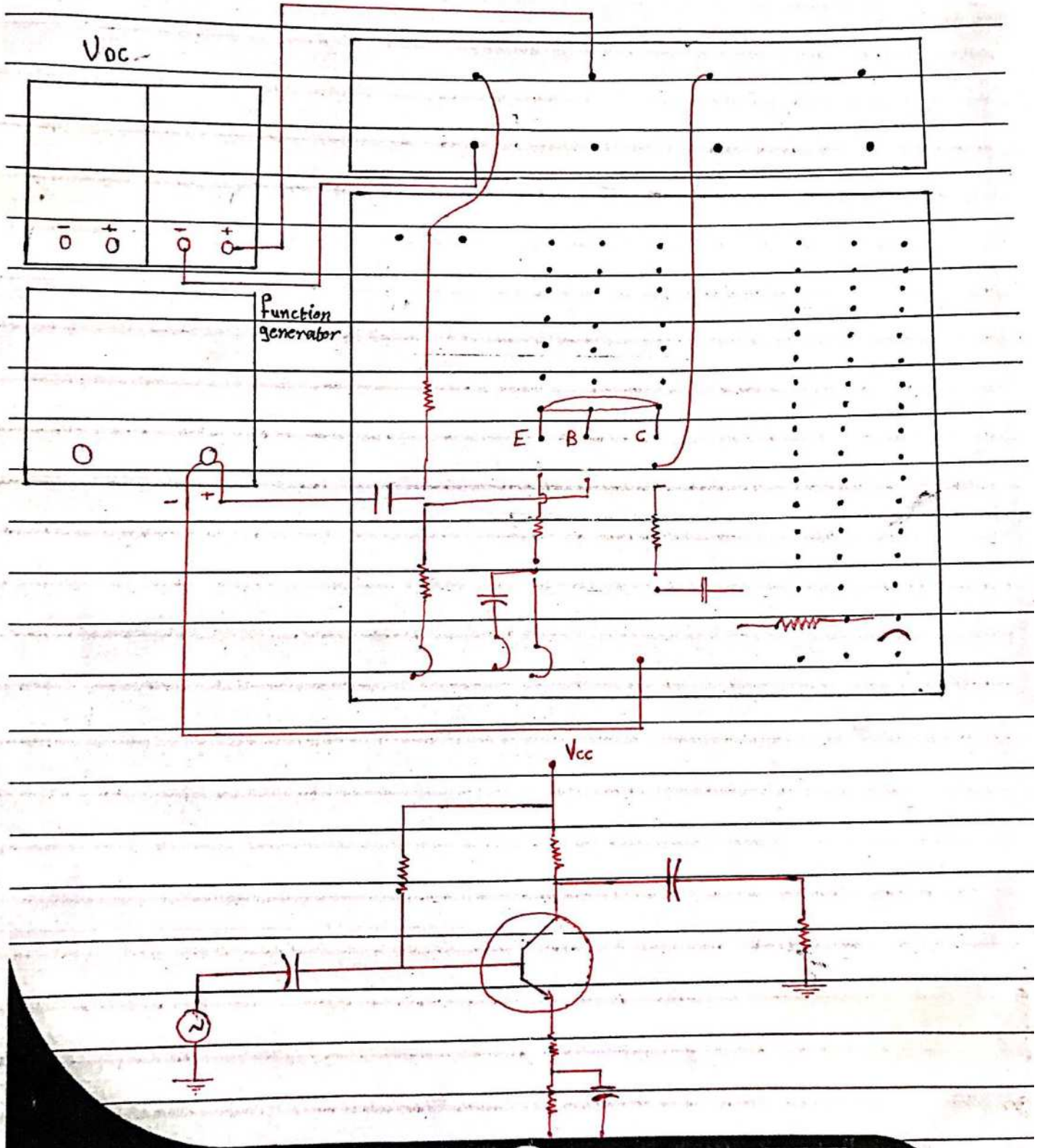
$A_{V_{NL}} = \frac{R_C}{R_{E1} + r_e}$ (Normal condition)

$A_{V_L} = \frac{R_C \,//\, R_L}{R_{E1} + r_e}$

$A_{V_{NL}} = \frac{R_C}{R_{E1} \,//\, R_{E2} + r_e}$ (No By Pass)

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Voltage Divider :



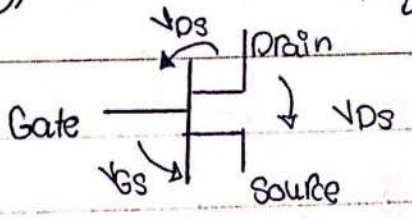
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*experiment -7-

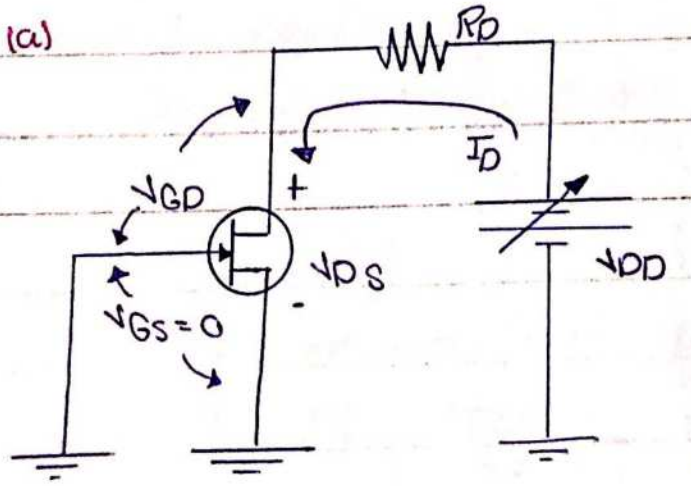
J-FET characteristics & The common-Drain Amplifier:

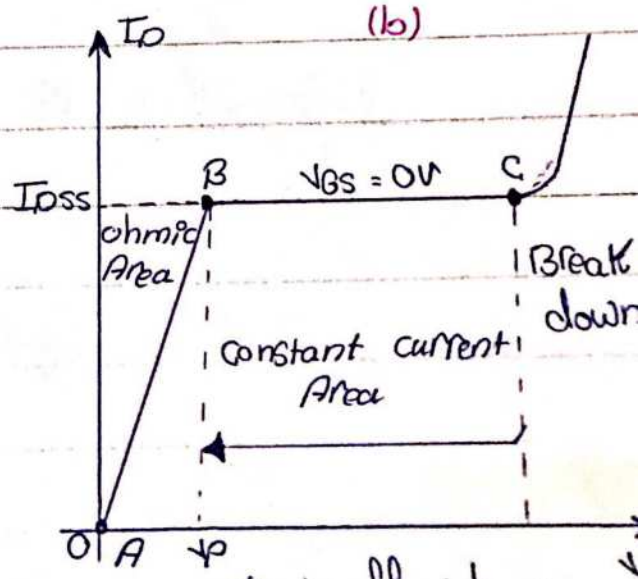
FET's

- J-FET
  Junction - field effect transistor
  - N-type
  - P-type
- MosFET
  metal-oxide semiconductor field effect transistor
  - enhancment
  - depletion



JFET characteristics circuit and curve.





JFET with VGS = 0 Volt and a vairable VDS (VDD)

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$I_D = 0$ when $V_{GS} = V_{GS(off)}$

And

$I_D = I_{DSS}$ when $V_{GS} = 0$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(on)}}\right)^2$$ JFET forward Tran-conductance

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

* Data sheets normally give the value of (g_m) measured at $V_{GS} = 0$ volt (g_{mo})

given the value of g_{mo} (Y_{fs}), g_m can be calculated

$$g_m = g_{mo}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)$$

if the value of (g_{mo}) is not available we can use the values of

$$g_{mo} = \frac{2 I_{DSS}}{|V_{GS(off)}|}$$

Useful formulas

Voltage gain (1) $A_V = \frac{V_{out}}{V_{in}}$ (2) $A_V = \frac{g_m R_S}{g_m R_S + 1}$

where $R_S = R_S \| R_L$

JFET dc gate-to-source cutoff voltage

(3) $V_{GS(off)} = \frac{2 I_{DSS}}{g_{mo}}$

Quiescent dc drain (source) current

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(4) $I_D = 2I_{DSS}\left[\frac{(R_S g_{mo}+1)-(2R_S g_{mo}+1)^{1/2}}{(R_S g_{mo})^2}\right]$

Quiescent dc gate-to source voltage

(5) $V_S = I_D R_S$

(6) $= -V_{GS}$ So that $V_G = 0$

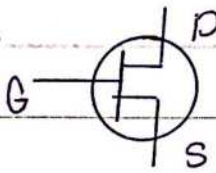
Quiescent dc drain-to-source voltage

(7) $V_{DS} = V_{DD} - I_D R_S$

JFET forward transconductance at Q point

(8) $g_m = g_{mo}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]$

* Example ..



$I_G = 0$, $I_D = I_S$

$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$

$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = g_{mo}\left(1 - \frac{V_{GS}}{V_p}\right)$

$g_{mo} = g_{mo}\left(1 - \frac{V_{GS}}{V_p}\right)$

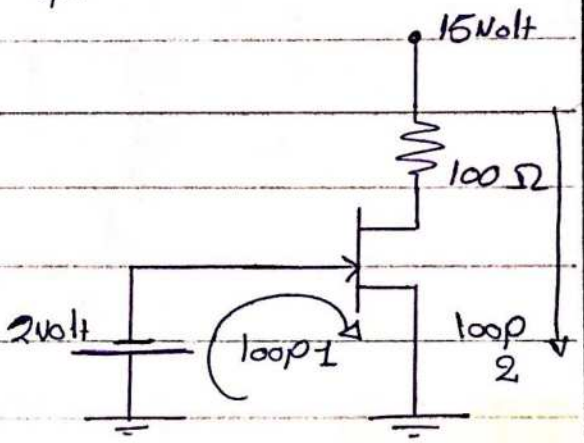
at loop one

$+2 + 1M * 16 + V_{GS} = 0$

$V_{GS} = -2$

$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$

$I_D = 8\left(1 - \frac{-2}{-4}\right) = 4\,mA$



$I_{DSS} = 8mA$

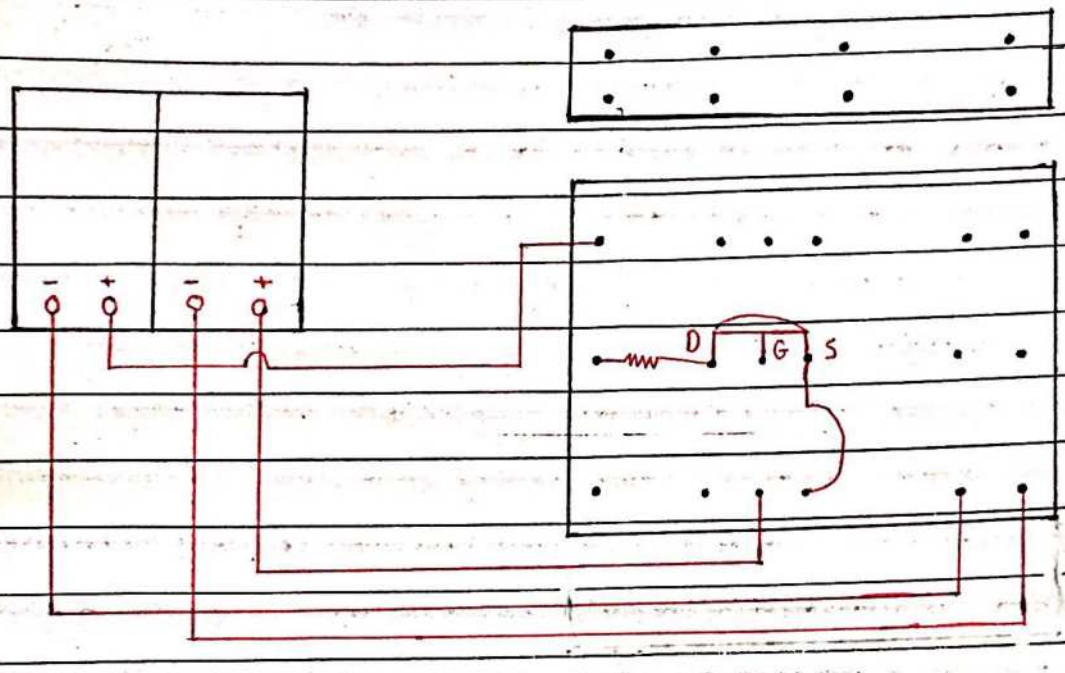
$V_p = -4$ Volt

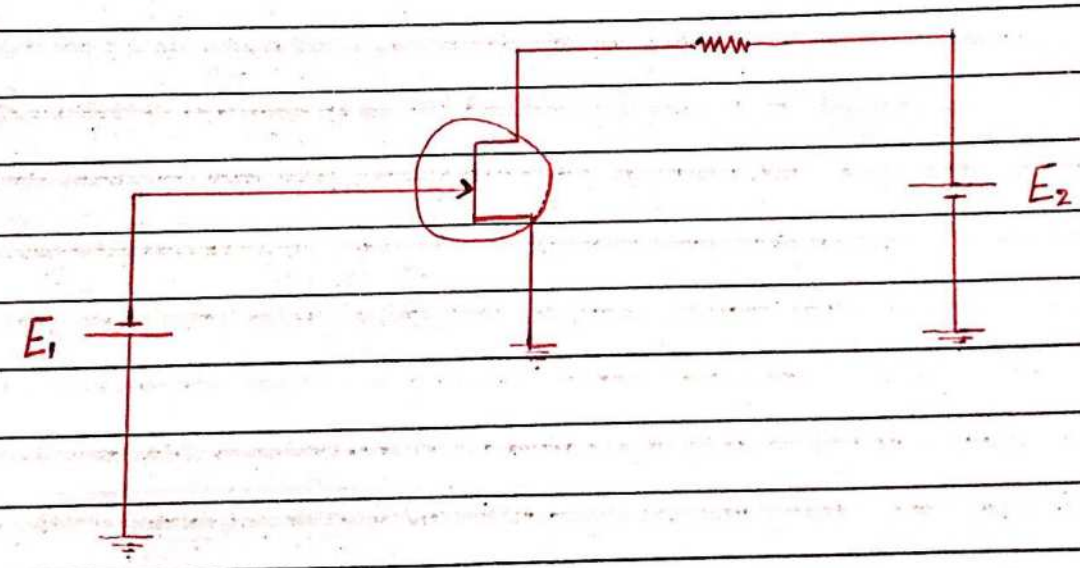
at loop 2

$-16 + 100I_S + V_{SD} = 0 \rightarrow V_{SD} = 14,6$ volt

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JFet characteristic :

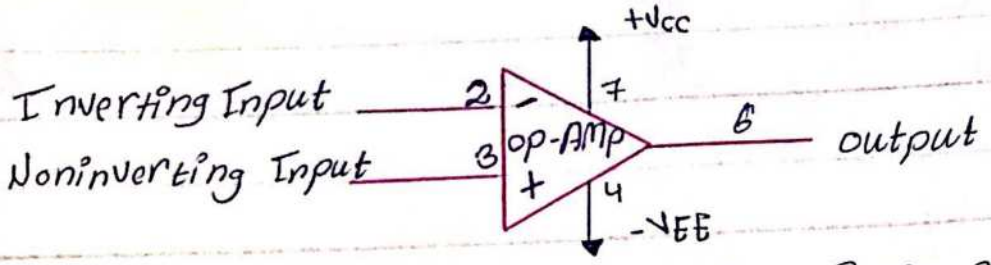




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Experiment -8-

OP-AMPS (operational amplifiers)



Op-Amp. Basic connections.

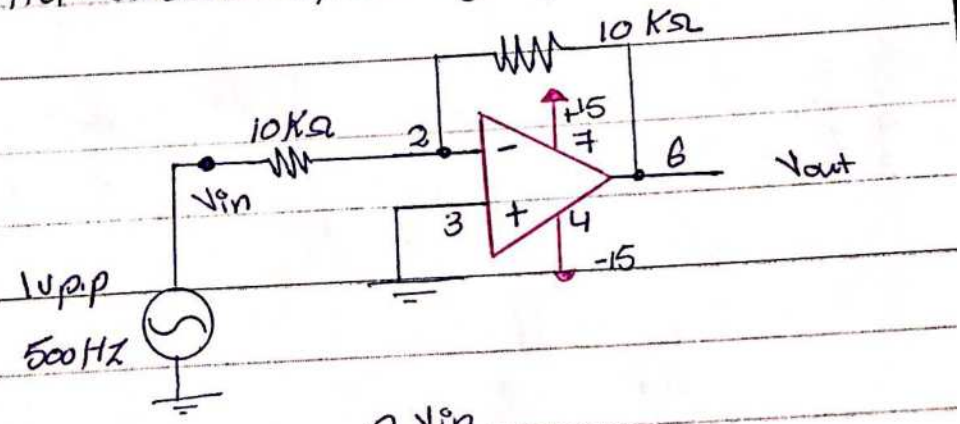
$V_o = A_v . V_i$

- Inverting Amplifier closed loop voltage gain.

$A_v = \frac{-R_f}{R_i}$

- Non Inverting Amplifier closed loop voltage gain.

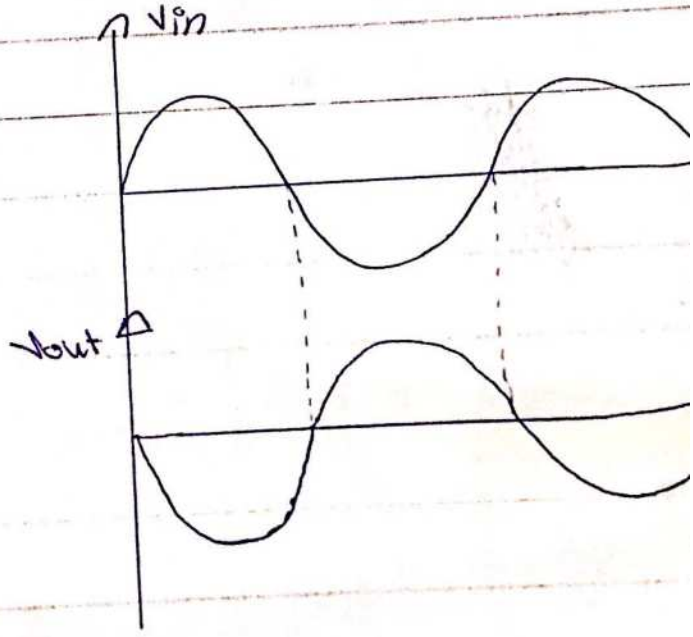
$A_v = 1 + \frac{R_F}{R_i}$



* Example:

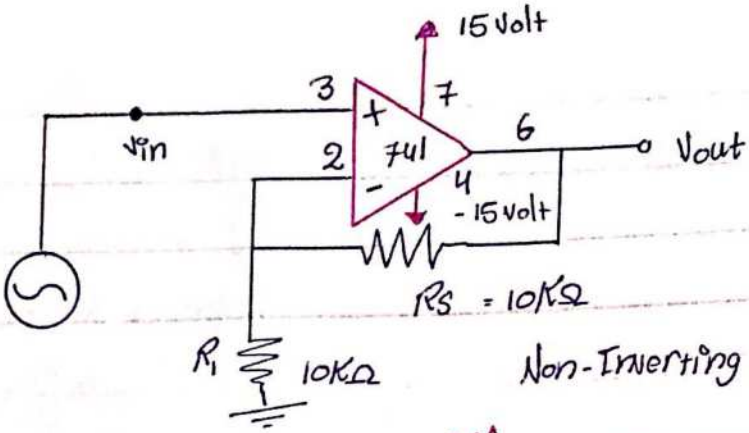
$A_v = \frac{-R_f}{R_i} = \frac{-10K}{10K} = -1$

o/p signal is always inverted with respect to i/p signal



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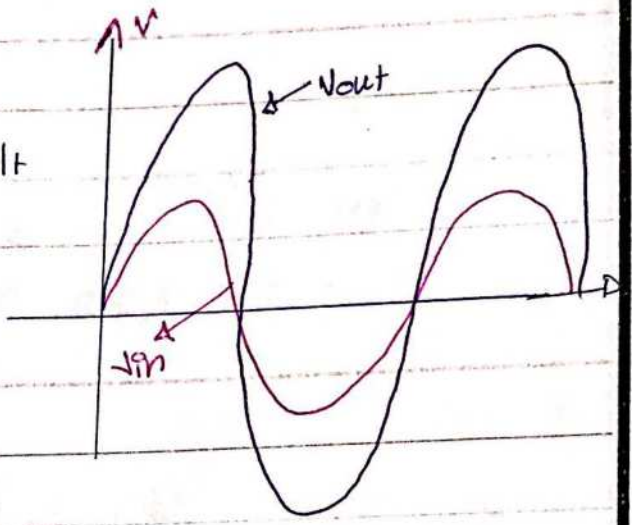
*Example



Non-Inverting Amplifier

$A_V = 1 + \frac{R_f}{R_1} = 2 \text{ volt}$

$V_{o/p} = A_V * V_{i/p} = 2 * 1 = 2 \text{ volt}$



o/p signal are always in phase

Useful formulas

Noninverting comparator output

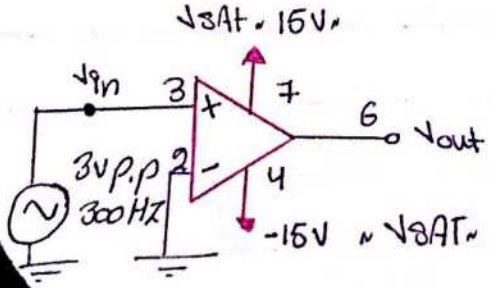
(1) $V_{out} = +V_{SAT}$ when $V_{in} > V_{REF}$

(2) $V_{out} = -V_{SAT}$ when $V_{in} > V_{REF}$

Inverting comparator output

(3) $V_{out} = +V_{SAT}$ when $V_{in} < V_{REF}$

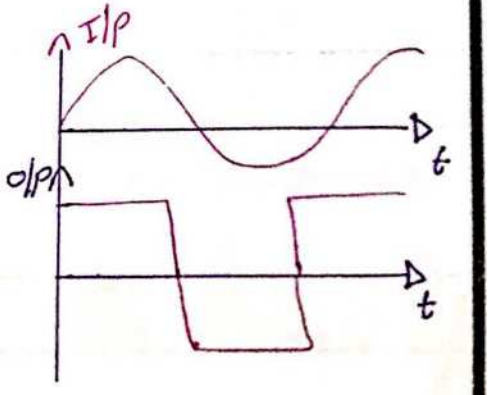
(4) $V_{out} = -V_{SAT}$ when $V_{in} > V_{REF}$



$V_{in} > V_{RF}$

$V_o = +15$

$A_V = \frac{V_o}{V_{i/p}} = \frac{15}{3} = 5$



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experiment -9-

OP-AMP Differentiator AND integrator.

* useful..

« Differentiator »

output voltage :

(1) $V_{out} = -RC\left(\frac{dV_{in}}{dt}\right)$

low-frequency response :

(2) $f = \frac{1}{2\pi RC}$

* when $f_{in} < f_c$ the circuit acts an a differentiator.
* when $f_{in} > f_c$ the circuit approaches an inverthing amplifier with a voltage gain of $-R_F/R_S$

« Integrator »

output voltage : (3) $V = -\frac{1}{R_I C}\int_0^t V_{in} \cdot dt$

low-frequency response :

(4) $f_c = \frac{1}{2\pi R_S C}$

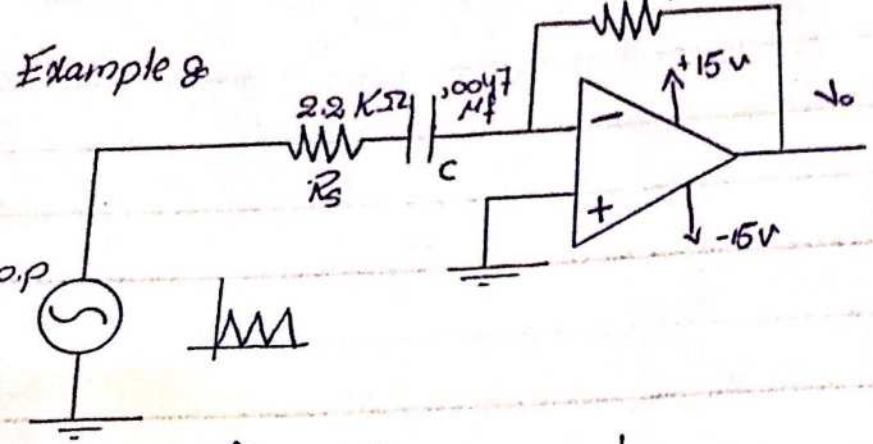
* when $f_{in} > f_c$ the circuit acts an integrator.
* when $f_{in} < f_c$ the circuit approaches an inverthing amplifier with a voltage gain of $-R_S/R_I$.

→ for minimum output offset due to input bias currents :

(5) $R_S = \frac{R_I R_S}{R_I + R_S}$

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Example 8



$V_o = -R_f C \left(\frac{dv_{in}}{dt}\right)$

$f_c = \frac{1}{2\pi R_s C} = \frac{1}{2\pi * 2.2K * .0047\mu} \simeq 15K \text{ Hz}$

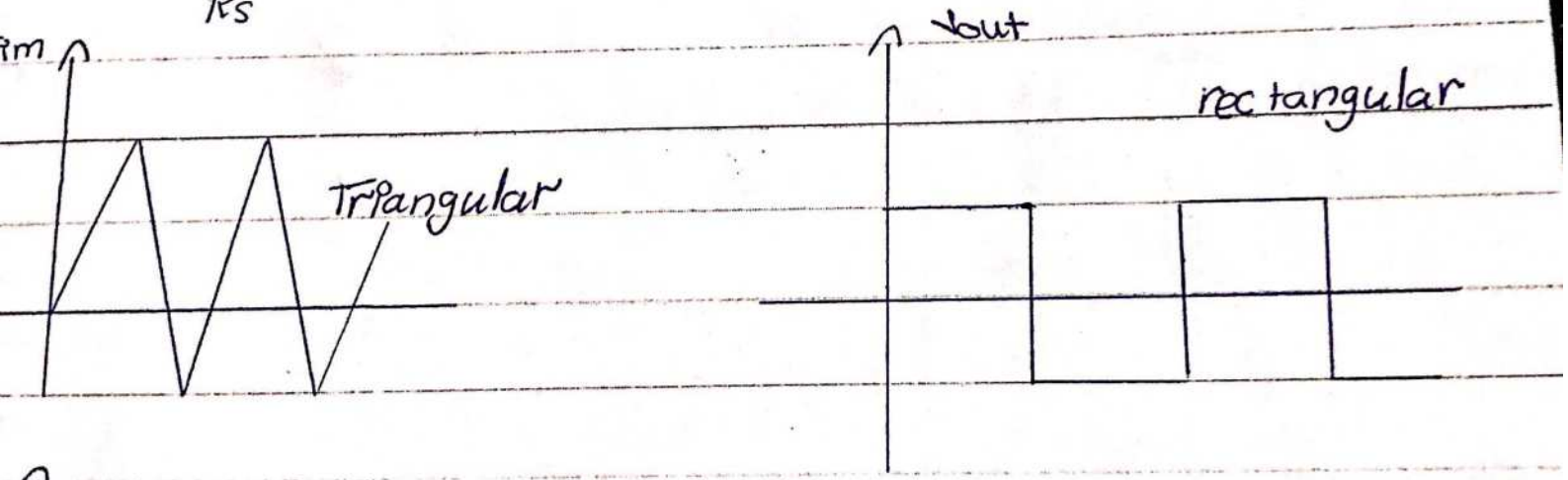
$f_c > f_{in}$ then $V_o = -R_F C \left(\frac{dv_{in}}{dt}\right)$

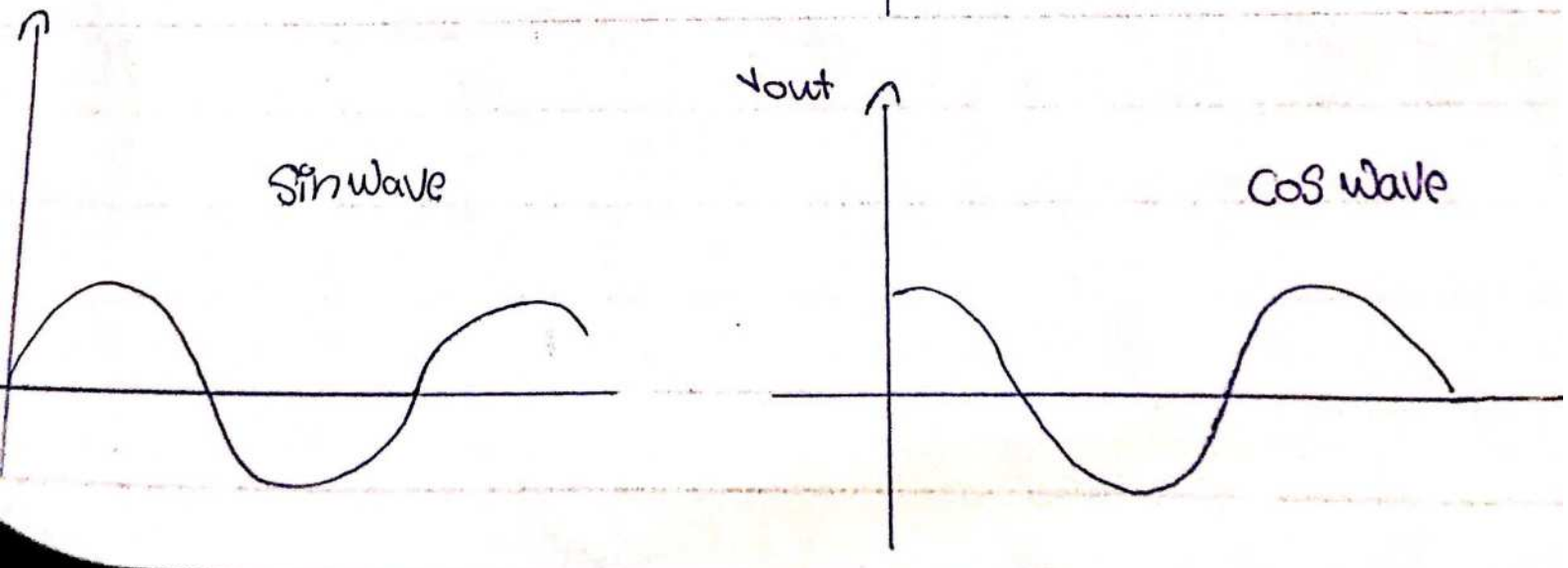
$V_o = -22K * 0.0047\mu (0.5)$

$V_o = -51.7\mu V$

if $f_{in} > f_c$

$A_V = \frac{-R_f}{R_s} = V_o = A_V . V_i$

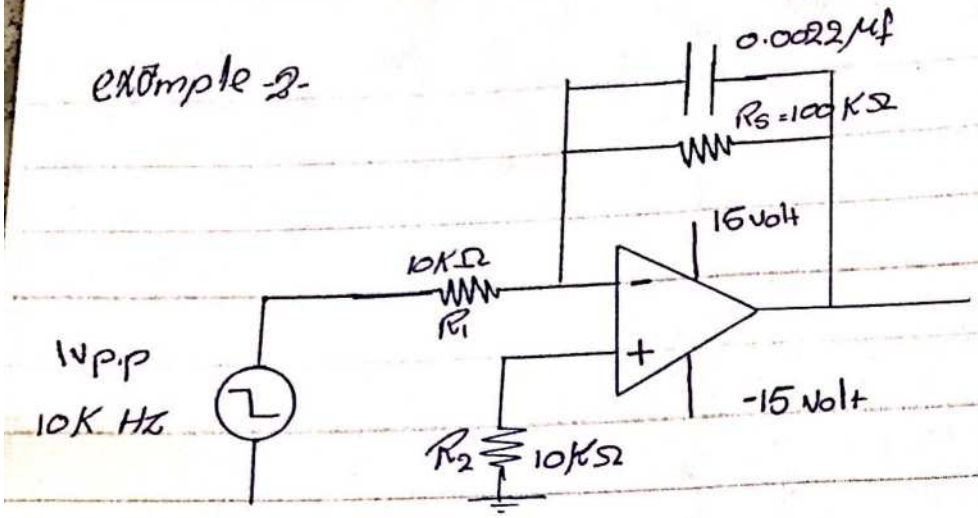




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example -2-



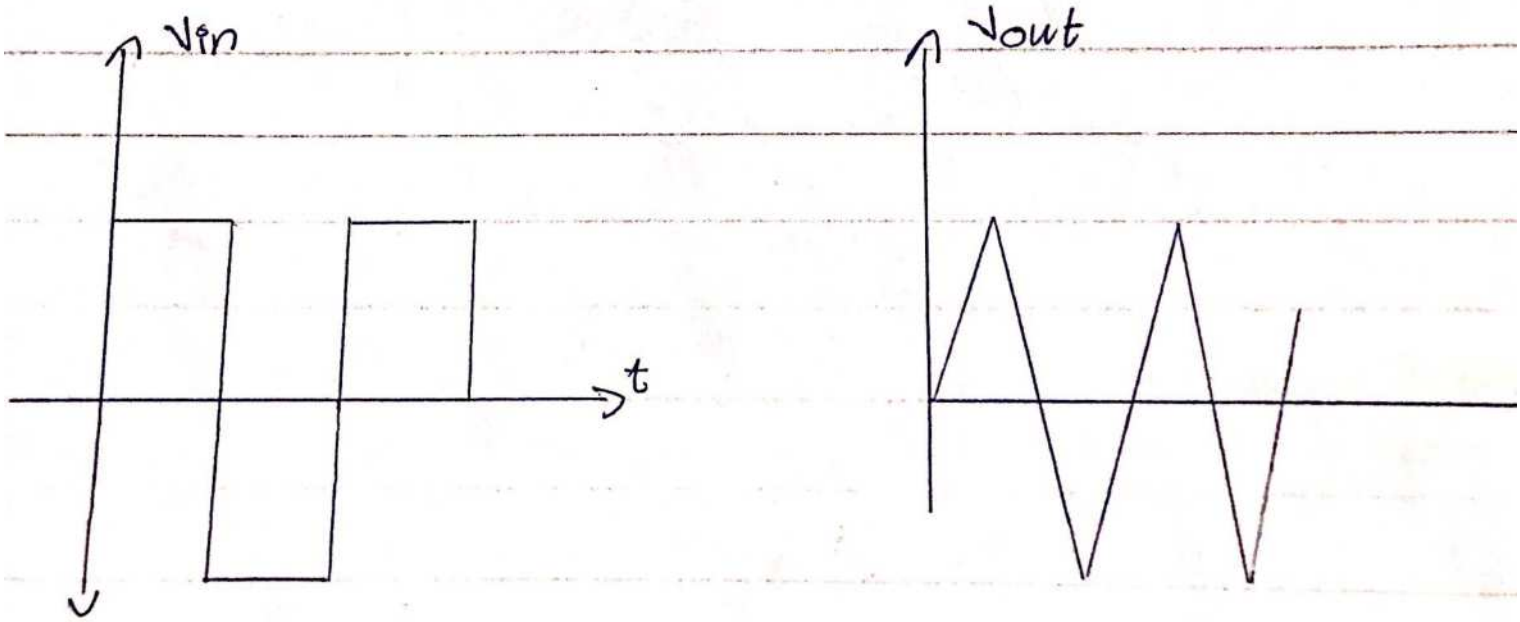
$$f_c = \frac{1}{2\pi R_S C} = \frac{1}{2\pi * 100K * 0.0022\mu} = 723.4 \text{ Hz}$$

$f_{in} > f_c$

$$V_o = \frac{1}{-R_1 C} \int_0^1 0.5 \, dt = \frac{-1}{10K * 0.0022\mu} * .5t = 22727.$$

if $f_{in} < f_o$ → $A_V = \frac{-100}{10} = -10$

$V_o = -10 * 0.5 = -5$ Volt



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